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Determinants of Market Value and Transfer Fee Deviations in European Football: A Two-Stage Quantitative Analysis of Player Pricing in the Top 5 Leagues

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Affidavit

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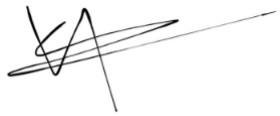
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Abstract

The European football transfer market moves billions of euros annually, yet the relationship between a player's Transfermarkt valuation and the fees actually paid in transactions remains underexplored. This study examines two related questions: which player characteristics determine market value, and which factors explain systematic deviations between realised transfer fees and those market value estimates. The present study focuses on permanent fee-paying transfers to clubs in the five major European leagues, the English Premier League, La Liga, Bundesliga, Serie A, and Ligue 1, across the summer windows of 2021, 2022, and 2023. This sample is comprised of 795 transfer observations, with Principal Component Analysis applied to performance data from FBref and two parallel sets of progressive Ordinary Least Squares regressions estimated on Transfermarkt market value and on the fee-market value gap respectively. The study aims to determine the mechanisms of price formation in the European transfer market and to identify the structural and player-level determinants of market value. The results may be used by both clubs and researchers in football economics, and more specifically, by clubs assessing value in transfer negotiations and by future researchers studying the informational content of crowd-sourced valuations in professional sports.

***Keywords**:* football economics, transfer fees, market value, player valuation, Principal Component Analysis, European football, Transfermarkt

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1. Introduction

In the summer of 2023, 20-year-old Jude Bellingham moved from Borussia Dortmund to Real Madrid for a reported €127 million. At the moment of his transfer, his Transfermarkt market value stood at approximately €120 million, which seems eerily close to the final fee Real Madrid agreed to pay. This individual transaction reflects a broader market reality: according to Poli et al. (2024), total global transfer spending reached €12.2 billion in 2023 alone, making the football transfer market one of the most economically significant labour markets in the world. Every summer, hundreds of these transactions take place across Europe's major leagues, each at a fee that is rarely identical to the player's published market value, and the question of why these two prices diverge remains remarkably under-studied.

In today's interconnected football industry, two distinct prices coexist for every professional player at every moment. The first is his market value, reevaluated continuously by different medias, companies, and communities and updated as performances evolve. The second is the realised transfer fee, paid by the buying club to the selling club at the moment of a transfer. The market value is itself a downstream product of a player's characteristics, it is what the crowd has already concluded about the player based on observable performance, age, position, and league context (Herm et al., 2014; Müller et al., 2017). The realised transfer fee is the outcome of a bilateral negotiation between buying and selling clubs, possibly drawing on that crowd valuation as a starting point but with additional considerations of urgency, contractual position, and bargaining leverage (Carmichael & Thomas, 1993). The two prices are therefore related but distinct, and the gap between them contains information about market dynamics that researchers have yet to fully grasp.

Academic research on football player valuation has long recognised that transfer fees are the product of multiple determinants, including performance, age, position, contract length, and league context. Hedonic pricing models pioneered by Dobson & Gerrard (1999) established the empirical template that has now dominated the literature for over two decades. Subsequent contributions have refined this approach using more granular performance metrics (McHale & Holmes, 2023), advanced econometric corrections for sample selectivity (Carmichael et al., 1999; Ruijg & van Ophem, 2015), and global sample frames (Poli et al., 2024). However, a critical gap persists: the dominant approach treats market value as just another regressor, entering it alongside player characteristics in a single equation predicting the realised fee. This produces high explanatory power but combines two distinct economic processes, the formation

of a market consensus about player quality, and the bilateral negotiation through which an actual transaction price is agreed. This research addresses this gap by examining these two processes separately, identifying which player characteristics firstly determine market value and which factors explain systematic deviations between realised transfer fees and those market value estimates.

The research questions addressing this gap are as follows:

RQ1: Which player characteristics determine the market value of professional footballers in Europe's Top 5 leagues?

RQ2: Which factors explain systematic deviations between realised transfer fees and Transfermarkt market value estimates?

The study focuses on permanent fee-paying transfers to clubs in the five major European leagues, the English Premier League, Spain's La Liga, Germany's Bundesliga, Italy's Serie A, and France's Ligue 1, across the three summer transfer windows of 2021, 2022, and 2023. The 2020 window was deliberately excluded to avoid the structural disruptions of the COVID-19 pandemic on transfer market activity. Performance data is sourced from FBref, while market values, transfer fees, and injury history are sourced from Transfermarkt. The analytical sample comprises 795 permanent fee-paying transfers, of which 722 involve unique players. Principal Component Analysis is applied to six per-90-minute performance metrics to address the multicollinearity inherent in football performance data, and two parallel sets of progressive Ordinary Least Squares regression models are then estimated, complemented by position-specific subgroup regressions and outlier-trimming robustness checks. By deconstructing the determinants of market value from the determinants of fee-market value deviations, this study provides a more complete account of price formation than the standard single-stage framework allows, with potential applications for clubs assessing market value in transfer negotiations and for researchers examining the informational content of crowd-sourced valuations in professional sports.

This thesis is structured as follows. The next section, Chapter 2, presents the literature review, which establishes the theoretical foundations of football as a labour market and surveys the empirical literature on transfer fees, market value, performance metrics, and emerging determinants. Chapter 3, the methodology section, describes the research design, data sources, sample construction, variable operationalisation, Principal Component Analysis, and the two-stage regression framework used to test the research questions. Chapter 4, the results section, reports the descriptive statistics and regression outputs for both sets of models. Chapter 5, the

discussion section, interprets the findings considering the prior literature and situates them within a cross-sport comparative framework. Finally, Chapter 6 concludes by summarising the contributions of the study and suggesting directions for future research.

2. Literature Review

This chapter reviews the academic literature on the determinants of football player transfer fees and market value, analyzing theoretical contributions from labor and sports economics as well as empirical studies of the European transfer market. The review examines the theoretical foundations of football as a labor market, the structure and regulation of the transfer system, and the role of performance metrics, market value, age, popularity, and injury as determinants of player valuation. This Literature Review aims to establish the conceptual framework for this study, identify the key variables and methodological approaches used in prior work, and identify the gaps this study addresses.

2.1 Football as a labour market: Theoretical foundations

Professional football today constitutes one of the most economically significant labour markets in the world. The 5 biggest leagues, all European, the English Premier League, Spain's La Liga, Germany's Bundesliga, Italy's Serie A, and France's Ligue 1, generated a combined revenue exceeding €15 billion in the 2022/23 season, with player transfers alone accounting for billions of euros in annual expenditure for clubs (Frick, 2007 ;Franceschi et al., 2024).

Yet the academic study of professional football as a labour market is not recent. Before our modern era of commercialisation, theoretical frameworks developed in earlier literature continue to inform contemporary research on player valuation and transfer fees.

One of the very first contributions to the economics of professional sport is widely attributed to Rottenberg (1956), who analysed the labour market in the United States for baseball players and established that competitive balance and player mobility (their ability to change franchise often in order to rebalance competition) are central to the functioning of professional sports leagues. Although Rottenberg's analysis concerned only baseball, its core insight, that players are financial assets whose allocation across employers generates market-clearing prices, proved directly applicable to other professional sports in different countries and settings. In Europe, Sloane (1971) extended this framework to football, arguing that English football clubs should be understood as utility maximisers rather than pure profit maximisers unlike firms in more conventional industries. Sloane (1971) argued that football clubs weigh sporting success, attendance, and league survival on the same scale as financial returns, which introduces specific structural features and goals into the football labour market that make it different from standard labour market models.

Only a few years later, Scully (1974), in an analysis of the Major League Baseball, studied the relationship between pay and performance. He demonstrated that player salaries could be

empirically linked to measures of player performance, and that the pre-free agency reserve clause system in baseball resulted in players being underpaid compared to their production of marginal revenue. Scully's hedonic approach, regressing compensation on objectively measurable performance statistics, established the methodological template that would subsequently be adopted across sports economics, including this present study of regression analysis to model transfer fees as a function of performance and other player characteristics in football.

Due to its fundamental differences with other more traditional labour industries, Kahn (2000) in his study characterised the sports industry as a "labour market laboratory." Kahn (2000) argued that the exceptional transparency of sports data, where performance statistics, compensation, and worker characteristics are publicly observable for all employees, made sports market much less victim to information asymmetries, which creates research opportunities that are unavailable in conventional industries. This is precisely what makes football an attractive setting for the present study: websites such as Transfermarkt and FBref provide detailed, systematic data on player performance, valuations, and transfer outcomes that would be near impossible, or much too costly, to assemble for workers in other sectors.

At the same time, the economic structure of professional football bears directly on player valuations. Rosen(1981), in his paper on the economics of superstars, demonstrated that in markets where consumers exhibit strong preferences for the most talented providers of a certain service, small differences in ability and talent can translate into disproportionately large differences in economic reward. The best performers earn vastly more than those ranked just below them, even when objective quality differences are marginal. Rosen's (1981) model predicts a highly skewed distribution of earnings, influenced mostly by the combination of imperfect substitutability between talent levels and the technology of joint consumption, whereby the most talented performer can serve a large market simultaneously. Professional football conforms closely to these two conditions: the global broadcasting and commercial revenues of elite clubs mean that elite players generate returns on a scale that make those of moderately talented competitors insignificant, producing the steeply convex transfer fee distribution observed empirically (Lucifora & Simmons, 2003; Frick, 2007).

The superstar theory was applied to European football by Lucifora & Simmons (2003), who found evidence of a superstar effect in the Italian Serie A wage determination. Using quantile regression, they showed that the returns to performance are significantly higher at the top of the wage distribution than at the median, consistent with the predictions of Rosen's (1981) model. Frick (2007), in a review of the football players' labour market across the five

major European leagues, further confirmed that the market exhibits features characteristic of both standard labour economics, with performance, age, and contract characteristics influencing compensation, and the superstar dynamics described by Rosen (1981), whereby a small number of elite players command fees and salaries that are much higher than those of their peers.

Together, these contributions establish the theoretical framework through which we can analyse the determinants of football player transfer fees and valuation. Football as a sport is a labour market governed by its own economic principles, in which player value reflects both objectively measurable productivity and market-wide demand for superstars. The following sections will build on the foundations we have established by examining how the literature has aimed to quantify and estimate the specific factors driving player valuation and transfer fees.

2.2 The transfer market: Structure, regulation, and the determinants of transfer fees

Unlike conventional labour markets or other sports labour markets such as basketball, in which workers change employers freely upon contract expiry, the football transfer market historically required the payment of a fee by the buying club to the player's current employer. This institutional feature of football has been the subject of multiple economic analyses since the early 1990s, seeking to explain the variation in observed transfer fees as a function of player, club, league and market characteristics.

The theoretical foundation for studying transfer fee determination was laid by Carmichael & Thomas (1993), who applied the Nash bargaining theory to the English Football League transfer market. In their framework, the transfer fee is conceptualised not as an exogenously determined price but as the outcome of a bilateral negotiation between the selling and buying clubs, both seeking to maximise its own utility function. Drawing on Sloane's (1971) specification of the football club as a utility maximiser, Carmichael & Thomas (1993) modelled the Nash Bargaining Solution as the fee that maximises the two clubs' utility relative to their respective disagreement result, the outcome each club would obtain in the absence of a transaction. Their analysis of the 1990–91 season found that the bargaining framework was broadly consistent with observed fees, confirming that both selling and buying club characteristics influence the agreed fee. To this day, this insight is more relevant than ever: transfer fees cannot be simply explained by a players' quality alone, since the bargaining positions of the clubs involved introduce additional variance that player variables cannot explain.

The apparent complexity of the transfer market was further examined by Carmichael et al. (1999), who addressed a significant methodological concern in the preceding literature.

Since only a minority of professional players are transferred in any given season, OLS (Ordinary Least Squares) estimation of transfer fee equations based solely on observed transactions is vulnerable to selection bias: the sample of transferred players in a transfer window is not a random draw from the population of all professional players. Carmichael et al. (1999) applied the Heckman two-step procedure to data from the 1993–94 English season to jointly model the probability of transfer and the fee conditional on transfer occurring. Their results demonstrated that player characteristics influence both the likelihood of transfer and the fee, and that ignoring selection led to biased estimates in prior work. The study remains methodologically instructive, though this present research focuses exclusively on players who were transferred, which naturally restricts our sample to observed transactions.

Dobson & Gerrard (1999) provided the most complete early analysis of transfer fee determinants in English professional football. Using a dataset of 1,350 transfers between June 1990 and August 1996, they estimated a hedonic price model in which the log-transformed transfer fee was regressed on player characteristics, selling-club characteristics, buying-club characteristics, and time effects. Their results confirmed that player performance, age, and positional characteristics are significant predictors of fees, and that the determination of fees differs markedly across market segments. Importantly, Dobson & Gerrard (1999) also documented the impact of the retain-and-transfer system and the subsequent transition toward greater player mobility after the Bosman ruling of December 1995, which established free agency for out-of-contract players moving between EU member states, constituted a structural break in the market.

The regulatory and financial implications of the Bosman ruling were examined in greater depth by Feess & Muehlheusser (2003), who analysed the consequences of three distinct transfer fee regimes: the pre-Bosman system, the post-Bosman system, and the proposed Monti system. Their theoretical analysis showed that the different regulatory environments influenced contract length, training investment, and effort provision, with implications for both club behaviour and the financial distribution of revenue between clubs and players. Feess & Muehlheusser (2003) concluded that the post-Bosman regime increased players' bargaining power and, consequently, their share of football's revenue, shifting the balance of power that Carmichael & Thomas (1993) had identified. This regulatory evolution is directly relevant to the present study, which examines transfers completed between 2021 and 2023, a period governed entirely by the post-Bosman framework in which clubs can only receive a transfer fee for currently contracted players.

More recent work has confirmed these earlier findings while addressing the limitations of single-country and single-period samples. Ruijg & van Ophem (2015) revisited the sample selectivity problem identified by Carmichael et al. (1999) and proposed an ordered probit correction method that they argued handles selectivity more efficiently within standard regression frameworks. Applied to a sample of English Premier League transfers, their results indicated that correcting for selectivity does not marginally alter the direction of coefficient estimates relative to simple OLS, though it improves precision, a finding that supports the robustness of the OLS specification adopted in this study. Richau et al. (2021) extended upon this to examine the role of club ownership structure in transfer fee determination, finding that foreign-majority investors (often richer) are willing to pay a premium relative to domestic investors, particularly in the upper quantiles of the fee distribution. Their results suggest that a clubs' financial characteristics, specifically the identity and resources of ownership, introduce additional variance in transfer fees beyond player characteristics alone, reminiscing the framework proposed by Carmichael & Thomas (1993).

Taken together, this body of work establishes several conclusions that inform the present study. Transfer fees are the result of a bargaining process in which both club and player characteristics are significant. The post-Bosman regulatory environment constrains the market to transfers of contracted players, which must be addressed in empirical work. OLS estimation of log-transformed fees remains the dominant approach in the literature and is methodologically relevant when the focus is restricted to observed transactions. And while early studies were confined to a single national context, mostly the English First Division/Premier League, the global nature of the transfer market, as documented by Frick (2007) and more recently by Franceschi et al. (2024), necessitates a multi-league, multi-season perspective. The following section looks at how player performances can be measured and used to explain variation in transfer fees.

2.3 Performance metrics as determinants of player value

Throughout the variables studied in academic literature, performance variables are the most analysed. Franceschi et al. (2024), in their review of 29 peer-reviewed empirical studies, identified performance as the dominant variable, appearing in every model variation reviewed across thirty years of research. Yet despite the common consensus on its relevance, the literature reveals a gap between the appeal of performance as a predictor of transfer fees and the challenge of using performance metrics appropriately, particularly given the positional differences in

football, the high intercorrelation among common performance indicators, and the growing availability of data.

The earliest empirical work on transfer fees studied what McHale & Holmes (2023) described as rudimentary performance metrics: goals scored, minutes played, and appearances. Dobson & Gerrard (1999) included goals and games played as regressors alongside age and club characteristics in their hedonic model of English transfer fees, finding that they were statistically significant but too small in explanatory power. Carmichael et al. (1999) also included playing record variables, appearances, goals, and experience, among the determinants of transfer probability and fee. The consistent finding across this early literature is that performance, measured by these basic metrics, is positively associated with transfer fees, but that it explains only a small share of transfer price variation when isolated.

A key limitation of basic performance metrics is their inability to capture the contributions that players make across different positions. Goals and assists are obviously biased toward attacking players and provide no information about defensive performance, passing quality, or pressing intensity. This problem has been widely acknowledged in the literature and has resulted in two complementary solutions: position-specific modelling and the development of more advanced, position-neutral metrics.

With the first approach, several studies have estimated position-specific models in which different performance variables are considered for different positional groups. Majewski (2016), analysing the most valuable football players, found that the significance of performance coefficients varied widely across forwards, midfielders, and defenders, with goal-related metrics being highly relevant for attacking players but largely irrelevant for defenders. Poli et al. (2022) similarly structured their econometric model of transfer fees with positional considerations in the variable selection. Thus, the relevance of positional heterogeneity to determination of transfer fees has been recognised across the literature, enabling the position-specific regressions adopted in the study.

On the second approach, recent research has invested in the development of more advanced and detailed performance metrics that are position-neutral and more granular than basic statistics. McHale & Holmes (2023) evaluated two systems in their study of 1,946 transfers across 69 leagues: Goal Impact Metrics (GIM), which assess a player's contribution to changes in expected goals, and expected Goals Plus Minus (xGPM) ratings, which measure how a player affects their team's net scoring chance differential. When included alongside basic metrics such as goals scored and minutes played in a linear regression framework, the advanced metrics improved the model fit significantly, with McHale & Holmes (2023) indicating that the

full model including advanced metrics and sofifa player ratings achieved an R^2 of approximately 0.55. Crucially, McHale & Holmes (2023) taught us that advanced performance metrics add predictive value beyond basic statistics, a finding that reinforces our use of per-90-minute statistics, progressive carries, and expected goals derived from FBref rather than simple seasonal total statistics.

The high degree of intercorrelation among performance metrics (where players who score more also tend to play more minutes, contribute more assists, and generate more shots) presents a further challenge for a regression-based analysis. Aydemir et al. (2021) applied dimension reduction techniques to player rankings across the top five European leagues, showing that the high-dimensional space of individual match statistics can be largely compressed into a small number of latent factors without any valuable loss of information. Their unsupervised framework, which ranked players across competitions using aligned per-90 minutes played statistics, provides methodological support for the Principal Component Analysis (PCA) approach adopted in the present study.

Kirschstein & Liebscher (2019), in their analysis of Bundesliga market values, found that a tight set of performance-derived variables, including goals per game, minutes played, and international appearances, explained a large share of valuation variance, and that their regression-based model compared favourably with the Transfermarkt crowd-sourced estimates. Müller et al. (2017) extended this work across the top five leagues over six seasons, developing a multilevel regression model where performance metrics served only as one of several layers of determinants, and they found that data-driven estimations outperformed crowd-sourced valuations particularly for much lesser-known players. Poli et al. (2024), in the most recent and comprehensive study of this genre, designed a global model across over 8,000 transfers from 2014 to 2024, achieving an R^2 of approximately 0.85. Their model was built using performance metrics, age, position, contract length, and league fixed effects, confirming that the relationship between performance and transfer fee is robust across different national contexts and market conditions.

Bell et al. (2024) raised an important issue in the performance-fee relationship by demonstrating that English players command a systematic premium in transfer fees compared to players with similar performances from other nationalities. Their analysis of Premier League transfers found that, controlling for performance metrics and age, English players were valued approximately 40% higher than comparable non-English players. This finding underscores that performance metrics, however well-calibrated, do not fully determine transfer fees; structural

and contextual factors, including league of origin, nationality premiums, and market conditions, introduce additional variance that performance alone cannot account for.

Taken together, the literature on performance metrics points to three conclusions that directly influence our methodology. First, performance is a necessary but not sufficient predictor of transfer fees: models relying only on performance typically achieve modest explanatory power, and the inclusion of market value, positional controls, and league fixed effects is required to achieve any plausible fit. Second, the specific choice of performance metrics matters: advanced, per-90-minute statistics outperform simple totals, and dimensionality reduction techniques such as PCA are effective tools for reducing the intercorrelation inherent in performance data mentioned previously. Third, position-adjustment is necessary: performance metrics have different significations depending on a player's role, and failure to account for this would produce biased results.

2.4 Market value, age, and player characteristics

Beyond performance metrics, the empirical literature has already identified a set of player-level characteristics that demonstrate consistent and significant influence on transfer fees independently of on-pitch output. Three variables stand out as the most influential: Transfermarkt market value, player age, and positional identity. This section examines each in turn, before addressing the role of nationality, physical attributes, and injury history as secondary determinants.

The relationship between Transfermarkt crowd-sourced market values and actual transfer fees has been a reoccurring subject of scrutiny. Herm et al. (2014) wrote the most early investigation of this relationship, finding that Transfermarkt market values explained approximately 90% of the variance in actual transfer fees in their Bundesliga sample, a very high level of correspondence meaning these crowd-sourced estimates could be used as a credible proxy for a players' true value. Herm et al. (2014) attributed this accuracy to Transfermarkt's "judge principle," where hierarchical information aggregation by designated expert users corrects for individual biases that would otherwise undermine the crowds' accuracy. Müller et al. (2017) proved these findings across the Top 5 leagues, while also showing that crowd-sourced values seem to always underestimate fees in the upper segment of the market, meaning that superstar premiums are not captured, or even, understood by community judgements and that models incorporating performance data can improve their predictions particularly by accounting for this. Poli et al. (2022) clarified that market value and transfer fee, while correlated, are distinct entities: market value reflects an estimate of what a

club might expect to receive for a player in case of a transfer, while a transfer fee is the actual price negotiated in a specific transaction and is influenced by urgency, negotiation skills and bargaining dynamics. This distinction needs to be made in this study, since we use Transfermarkt market value as a regressor rather than a dependent variable. Kirschstein & Liebscher's (2019) Bundesliga study reported that Transfermarkt maintained a correlation of 93% with realised transfer fees, further justifying its inclusion as a control variable in our regression frameworks.

Franceschi et al. (2024) found that age or age squared was included in 74 of the 111 regression specifications they examined, making it the single most frequently tested variable. The dominant conclusion is an inverted-U relationship. A player's value rises during their early career years as experience accumulates game time, potential is at its highest along with market value speculation. It reaches a peak in the mid-to-late twenties, and declines thereafter as physical capacity begins to diminish. Müller et al. (2017) confirmed this pattern across the top five leagues, estimating peak value at approximately 26 to 28 years old. Metelski (2021), analysing the Polish Ekstraklasa transfers, found that the highest transfer fees in the league were for players aged 21 to 24, consistent with the view that clubs value future potential heavily alongside current performance and are therefore willing to pay a premium for young players, whether it to be enjoy these players at their peak in the future, capitalize now and avoid paying more in the future, or fear of missing out on talent. Khalife et al. (2025) extended this analysis using Gradient Boosting machine learning models segmented by career stage, finding that early career valuations are heavily influenced by potential and development trajectory, while late-career valuations depend more on current performance output. This has many important implications for the present study: our 2021–2023 sample period captures a market in which clubs have been paying historically high fees for young players, and age is therefore expected to have a positive effect on fees among transferred players.

Beyond age, several other influential player-level characteristics have been identified. Szymanski (2000) found no systematic evidence of discrimination in transfer prices in the English Premier League, concluding that the market was somewhat efficient in pricing player talent irrespective of ethnicity or nationality. However, as mentioned in Section 2.3, Bell et al. (2024) found a persistent English player premium in the Premier League transfer market, suggesting that nationality effects in transfer markets are real but operate at a league-level rather than at an individual level. Lago-Peñas et al. (2019) examined the relationship between players migration patterns and national team selection and performance, finding that the players who move to higher-ranked leagues than their national ones tend to be precisely those identified by

their national teams as performing at an elite level, a selection effect that reinforces the view that transfer market decisions respond to quality signals, with national origin being of these signals.

Müller et al. (2017) found that position also influences market value, with midfielders constituting the largest share of their sample and attackers commanding disproportionately high values relative to defenders. Our position-specific models confirm the pattern documented in the literature by Franceschi et al. (2024): controlling for all other factors, forwards are associated with a significant transfer fee premium. Finally, Follert & Gleißner (2024) proposed that transfer fees should theoretically reflect the discounted expected future cash flows a player generates for the buying club, through merchandising revenue, club performance and additional performance revenue.

2.5 Popularity, media, visibility, and the superstar premium

The literature reviewed in the preceding sections treats player value mostly as a function of objective, on-pitch contributions. However, with the growing influence of social media in society and sport as an industry, a player's public profile and visibility, independent of their sporting output can exert a significant influence on their valuation.

The theoretical case for popularity as a valuation driver was studied most rigorously by Hofmann et al. (2021), who developed a framework linking performance-based and popularity-based attributes to players market values through brand image. Drawing on Keller's (1993) brand knowledge framework and Rosen's (1981) superstar theory, Hofmann et al. (2021) argued that football players are human brands whose market values reflect both their sporting talent and their accumulated consumption capital, the degree to which fans and the public have invested attention and emotional engagement in a player over time. By combining primary survey data with secondary market value data for 316 Bundesliga players, they found that both performance attributes and popularity attributes significantly influenced brand image, and that brand image then had a significant positive effect on market value ($R^2 = 0.55$ for the brand image construct, $R^2 = 0.15$ for the market value model).

Majewski (2021), in his analysis of the 150 most valuable attacking players, accounted for the popularity premium through his Brand Importance Ratio (BIR), defined as the ratio of a player's annual earnings to their market value. Majewski (2021) found that several elite players, including those appearing in Table 1 of his study, exhibited BIR patterns consistent with a significant brand effect, additional to their performance-based valuation, confirming that

for the most visible players in the sport, popularity functions as an independent factor of economic value that standard regression models will underestimate.

Prockl & Frick (2018), in a study of the Transfermarkt.de community across ten seasons of Major League Soccer data, found that crowd-sourced market value estimates, which themselves include community members' assessments of popularity alongside performance, functioned as excellent proxies for player salaries, with correlations exceeding those achievable from objective performance statistics alone. Their finding reinforces the view that popularity is not merely a premium reserved to football superstar, but a factor that shapes market valuations across every player.

The most methodologically rigorous recent contribution in this section is Malagón-Selma et al. (2023), who proposed to construct popularity indicators for football players using Google Trends. Unlike raw search volume, Google Trends provides normalised, comparable indices of Google's search engine interest that can be measured over time. Malagón-Selma et al. (2023) constructed six distinct popularity indicators and tested their predictive validity for Transfermarkt market values across the top five leagues using linear regression, random forest, and gradient boosting models. Their results consistently showed that popularity indicators improved model fit and reduced transfer fee prediction deviations across all three methods. While this study does not incorporate a social media popularity variable as an independent regressor (a limitation that is discussed further in Section 5) the framework developed by Malagón-Selma et al. (2023) provides a clear blueprint for how such a popularity variable could be constructed and incorporated in future work building on these findings.

Taken together, the literature on popularity and superstar premiums makes clear that a complete model of transfer fee determination must grapple with the tension between objective performance and subjective public perception. However, in our favour, Transfermarkt community members do not evaluate players on performance alone; they form judgements considering a player's global profile, their national team status, their club prestige, and their commercial visibility. This means that the market value variable, as used in the present study, already partially captures the popularity premium mentioned, though it does so indirectly and imprecisely.

2.6 Research gaps and positioning of the present study

Despite the fact football player transfer fees and market valuation have been covered for three decades, the literature on the topic still contains significant gaps which we aim to address.

The field suffers from three interconnected limitations that motivate the present study. First, the literature is heavily focused on the English leagues and on data from the early 2000s to mid-2010s, with few studies covering the post-2019 transfer market and the specific dynamics of the 2021–2023 period, a window shaped by the economic disruptions of the COVID-19 pandemic, record-breaking transfer activity, and the accelerated globalisation of the player market (Poli et al., 2024). Franceschi et al. (2024) explicitly identify the need for research on more recent seasons across a multi-league sample in order to capture and compare league effects. Our study addresses this gap directly by analysing 795 permanent transfers from all five major European leagues across the 2021–2023 window, with season and league fixed effects incorporated throughout.

Second, the multicollinearity problem in performance data has been widely acknowledged in the literature, Franceschi et al. (2024) dedicate a specific discussion to the issue in their paper, noting that the abundance of correlated performance metrics is at risk of producing uninterpretable coefficients, but still this has rarely been addressed in transfer fee regression studies. The most common approach in the current literature is simply to take into account a small number of performance variables individually and accepting the multicollinearity as a limitation. Kirschstein & Liebscher (2019) applied robust PCA to Bundesliga market value data and demonstrated that dimensionality reduction gives more stable and interpretable models, but this approach has not been widely adopted yet. We apply PCA to the full set of FBref performance statistics, in order to reduce the correlated performance space to three orthogonal principal components before entering them into the regression, which will respond directly to the gap Franceschi et al. (2024) identified by applying a similar method to Kirschstein & Liebscher (2019) and Aydemir et al. (2021).

Third, and finally, position-adjusted analysis remains underdeveloped in the literature. While most studies have acknowledged that performance coefficients vary across positional groups, the systematic comparison of how performance translates into transfer fees differently for defenders, midfielders, and forwards, has not been consistently analysed across the empirical literature. Khalife et al. (2025) demonstrate using machine learning that the most influential features vary substantially by position, with shooting metrics dominating for attackers, passing metrics for midfielders, and defensive skills for defenders, reinforcing the need for position-focused performance analysis. We address this by estimating both a pooled model with position dummies and interaction terms alongside three position-specific subgroup regressions, providing a direct empirical test of whether the performance-fee relationship is homogeneous across positions.

3. Methodology

This chapter describes the research design, data collection procedures, sample construction criteria, variable operationalisation, and analytical methods employed in this study. Each decision is explained and, where relevant, justified with reference to established practice in the empirical literature on football player valuation. Detailed regression outputs and diagnostic plots are provided in the Appendix.

3.1 Participants and the design

This study adopts a quantitative, cross-sectional research design based on secondary data collected from publicly available sources. The underlying assumption of the study is that the determinants of football player transfer fees can be identified and estimated through systematic observation and statistical analysis (Kahn, 2000). This approach is consistent with the dominant method in the sports economics literature, where OLS regression of log-transformed transfer fees on player and market characteristics has constituted the standard empirical framework since the contributions of Dobson & Gerrard (1999) and Carmichael et al. (1999), and has since been adopted in all major recent contributions (Müller et al., 2017; Poli et al., 2022, 2024;McHale & Holmes, 2023).

The choice of using secondary data design is primarily due to the unique transparency of professional football as a research setting. As Kahn (2000) observed, there is no other industry in which the identity, performance record, physical characteristics, and compensation of every individual worker are publicly and systematically observable across time and region. Transfermarkt and FBref together provide the most comprehensive publicly available sources of transfer fee, market valuation, and performance data for European professional football, and have been adopted as primary data sources in a substantial share of the empirical literature reviewed in Chapter 2 (Herm et al., 2014;Müller et al., 2017; Kirschstein & Liebscher, 2019;McHale & Holmes, 2023).The use of these sources is therefore methodologically appropriate, due to these websites data being regularly updated.

This study covers the summer transfer windows of 2021, 2022, and 2023, corresponding to the three consecutive windows preceding the 2021/22, 2022/23, and 2023/24 seasons respectively. In the dataset, these windows are identified by the FBref convention of labelling each season by its end year, such that the summers of 2021, 2022, and 2023 appear as season years 2022, 2023, and 2024 respectively. This temporal window was chosen for two reasons. First, it avoids the distorting effects of the COVID-19 pandemic on transfer market activity. The summer 2020 window was structurally unusual, delayed by approximately two months,

compressed into a shorter negotiating period, a significant rise in loans of players instead of transfers and conducted under conditions of financial uncertainty across European clubs, causing some players to be transferred for fees under their market value. Its inclusion risked introducing systematic bias into fee levels that would not reflect the determinants of value under normal market conditions. Beginning the sample in summer 2021 ensures all observations reflect a fully normalised, post-pandemic transfer market. Second, the three windows covered, 2021, 2022, and 2023, encompass meaningfully distinct market cycles, including the post-COVID recovery phase, a period of record-breaking fee inflation, and the most active summer window in European football history to that point (Poli et al., 2024), providing adequate variation in market conditions across seasons for this study to be an accurate representation of market's evolution.

The geographic scope is restricted to the five major European professional football leagues: the English Premier League, Spain's La Liga, Germany's Bundesliga, Italy's Serie A, and France's Ligue 1. These leagues are collectively the most commercially significant and well-documented, as they account for the largest share of global transfer fee expenditure (Frick, 2007; Franceschi et al., 2024). It should be noted that the study includes transfers where the buying club is in one of the five leagues regardless of the selling club's origin, which may be located in any domestic or international competition worldwide. Restricting the sample to these five receiving leagues ensures comparability across observations and allows league effects to be retained, as described by Bell et al. (2024) who documented major and persistent inter-league differences in transfer fee levels that cannot be attributed to player characteristics alone.

Only summer window permanent transfers involving a positive disclosed fee are included in the sample. First, winter window transfers are excluded because they only represent a small number of clubs, players, and could distort the data as winter transfers are usually done out of urgency whether it be due to performance issues, or injuries. Loan transfers are excluded as well because they do not involve the payment of a permanent transfer fee and therefore represent a structurally different type of transaction. Free transfers, where a player's contract has expired and no compensation is owed to the selling club, are excluded because the log transformation of a zero fee is undefined, and because the economic determinants of a free transfer differ fundamentally from those of a fee-paying one. Similarly, transfers for which the fee was not publicly disclosed are excluded, as the absence of a reported fee prevents the construction of the dependent variable, but this won't account for a large amount of missing data as unreported fees are usually for small scale transfer of players. Goalkeepers are excluded from the sample due to their performance metrics being incomparable to outfield players.

Goalkeeping performance metrics (saves, clean sheets, distribution) have no meaningful relationship to the performance variables used in the Principal Component Analysis described in Section 3.5, and their inclusion would distort the dimensionality reduction procedure. This exclusion is consistent with the approach adopted by Müller et al. (2017) and McHale & Holmes (2023). Finally, a minimum playing time of 900 minutes, equivalent to ten full 90-minute matches, is applied to ensure that performance statistics are statistically representative of a player's ability level. Players below this threshold may have performance statistics that are too highly volatile and unrepresentative of their true quality.

After applying these filters, our final analytical sample is comprised of 795 player-season observations drawn from all five leagues across the three summer windows of 2021, 2022, and 2023. The sample involves 722 distinct players, with some players appearing in multiple seasons due to having been transferred more than once during the study window. The unit of observation is accordingly the player-season transfer event rather than the individual player, a structure consistent with Poli et al. (2024) and McHale & Holmes (2023). The distribution of observations across leagues and seasons is presented in Table 1.

Table 1. Sample distribution by league and season.

League	2021	2022	2023	Total
Bundesliga	36	45	54	135
La Liga	30	37	45	112
Ligue 1	45	55	65	165
Premier League	54	71	59	184
Serie A	53	64	82	199
Total	**218**	**272**	**305**	**795**

Note: Years correspond to summer transfer windows. Source: Author's own calculations based on Transfermarkt and FBref data.

Transfer fees in the sample range from €70,000 to €127 million, with a mean of approximately €14.3 million and a median of €8.0 million, demonstrating the right-skewed distribution that justifies the logarithmic transformation of the dependent variable discussed in Section 3.4. The mean player age is 24.8 years, with a range of 17 to 37. Forwards constitute the largest positional group (302 observations, 38.0%), followed by defenders (275, 34.6%) and midfielders (218, 27.4%).

3.2 Data sources and collection

The empirical analysis draws on three distinct data streams; each collected from a separate source and merged into one analysable dataset. All data was collected programmatically in RStudio (Posit Team, 2026) using R (R Core Team, 2025) and the worldfootballR package (Zivkovic, 2024), which provides a set of functions for accessing publicly available football data from Transfermarkt (2024) and FBref (2024). The collection process was conducted between March and April 2026, with rate-limiting protections implemented throughout to ensure a responsible and sustainable access to these platforms.

Transfer fees and player information were sourced from Transfermarkt, the German football database widely recognised as the most comprehensive publicly available source of player transfer records and market valuations in European professional football (Herm et al., 2014; Müller et al., 2017). Transfer data was collected using the `tm_league_team_urls()` and `tm_team_transfers()` functions, which retrieve the team URLs for each league-season combination and scrape the transfer records associated with each club. Data was collected for all five leagues across the three seasons 2022–2024 in the FBref labelling convention, corresponding to the summer windows of 2021, 2022, and 2023. To avoid server overload and respect Transfermarkt's rate-limiting policies, `Sys.sleep()` delays of between 5 and 20 seconds were inserted between requests, and a `tryCatch()` error-handling wrapper was applied to each loop iteration so that a failed individual request would not terminate the entire collection process. The raw transfer dataset contained all transfer types across all available seasons; filtering to permanent transfers with a disclosed positive fee, restricted to the three study windows, was applied at the cleaning stage described in Section 3.3.

Player market values were also sourced from Transfermarkt, using the `tm_player_market_values()` function. Market values were collected for all five leagues for each of the three season years 2022, 2023, and 2024 in the FBref labelling convention, yielding approximately 14,300 observations after retaining the most recent market value record per player per season. The first collection window was particularly challenging due to Transfermarkt's rate-limiting behaviour, which blocked requests when all five leagues were queried simultaneously; this was resolved by collecting the data country-by-country in separate requests with pauses. Market values on Transfermarkt represent crowd-sourced assessments of the amount a club would be expected to pay to sign a player under normal market conditions, and have been shown to correlate with realised transfer fees at approximately 0.93

in the German Bundesliga (Kirschstein & Liebscher, 2019) and to explain up to 90% of transfer fee variance in regression models (Herm et al., 2014).

Player performance statistics were sourced from FBref, a publicly available football statistics platform operated by Sports Reference LLC that draws on data provided by StatsBomb. FBref provides a comprehensive report of statistics for all players appearing in the Top 5 European leagues, updated on a rolling basis throughout each season. Performance data was collected using the `load_fb_big5_advanced_season_stats()` function with `stat_type = "standard"` and `team_or_player = "player"`, which retrieves per-player, per-season statistics for all five leagues simultaneously. Data was collected for season end years 2022, 2023, and 2024, corresponding to the three summer windows under study. The standard statistics dataset includes minutes played, goals, assists, expected goals (xG), non-penalty expected goals (npxG), progressive carries, progressive passes, and a range of additional metrics. For season end year 2024, the FBref dataset contained approximately 2,852 rows per season, yielding approximately 11,500 rows across the three seasons prior to filtering. Unlike the Transfermarkt collection, FBref scraping via the `load_fb_big5_advanced_season_stats()` function did not encounter rate-limiting issues, as the function retrieves pre-compiled statistical data directly from the FBref platform, making it much faster.

Injury history was sourced from Transfermarkt using the `tm_player_injury_history()` function, which retrieves the full recorded injury history for an individual player given their Transfermarkt profile URL. Transfermarkt player profile URLs were retrieved directly from the transfer and market value data collected via worldfootballR, which returns player profile URLs as part of the standard output of the `tm_league_team_urls()` and `tm_player_market_values()` functions, and were joined to the analytical dataset using standardised player names as the matching key. Injuries were scraped individually for each of the 795 players in the final sample in a sequential loop, with a `Sys.sleep(5)` delay between each request and a checkpoint save every 50 players to prevent data loss in the event of a connection failure. The resulting injury dataset was aggregated to the player-season level by summing all recorded injury days within each relevant season window, producing a single `injury_days` variable per observation. Of the 795 players in the final sample, 490 (61.6%) had at least one recorded injury day in the relevant season, while 305 (38.4%) had no recorded injuries. Players who had no injury records returned by the scraper were assigned a value of zero, consistent with the reasonable assumption that the

absence of a recorded injury indicates an injury-free period rather than missing data, given Transfermarkt's comprehensive injury tracking coverage for Top 5 league players.

The three data streams, transfer fees, market values, performance statistics, and injury history, were merged into a single master dataset using player name and season year as join keys. The merge procedure and subsequent cleaning steps are described in Section 3.3.

3.3 Sample construction and cleaning

The construction of the final analysable dataset involved merging the three data streams described in Section 3.2 into a single master file, through cleaning and filtering steps, and resolving data quality issues that came during the join procedures.

The first step was the standardisation of player names across the three datasets. Because each source uses a different naming convention, FBref uses romanised full names, Transfermarkt uses localised names, and the player reference file contains a mix of both; direct string matching is unreliable without prior normalisation. A cleaning function was applied to all three datasets that performed Latin-ASCII transliteration to remove marks and non-standard characters, converted all names to lowercase, and removed leading, trailing, and internal excess whitespace. The resulting standardised name field `(player_clean)` served as the primary join key for all future merges, complemented by `season_year` to ensure that each join matched players to the correct seasonal record.

The market value dataset required pre-processing before merging. Transfermarkt reports market values at irregular intervals throughout the season. To obtain a single representative market value per player per season, the most recently recorded valuation within each season year was retained for each player, using a grouped slice operation on the date variable. The resulting player-season market value records were then joined to the FBref performance dataset using a shared `player_id` field, which was itself obtained by joining FBref player names to the Transfermarkt player reference file. This two-step join procedure was necessary because FBref and Transfermarkt do not share a common player identifier.

The name-based join produced a relationship warning for a subset of observations, caused by players sharing identical standardised names (a known challenge in football data merging) where common names such as "Luis Garcia" or "David Silva" appear in multiple rows of the reference file. Following the join, the master dataset contained 6,533 rows against 6,018 unique player-squad-season combinations, indicating 515 duplicate rows attributable to this issue. Duplicates were resolved by retaining, for each player-squad-season group, the row associated with the highest market value.

Transfer fee records were joined to the merged FBref-market value dataset using `player_clean` and `season_year` as keys. When a player had multiple transfer records in the same season, the record with the highest fee was retained. The resulting master dataset contained 6,018 unique player-season observations across all five leagues and four season years, of which 1,066 had a non-zero, non-missing transfer fee. The remaining 4,952 observations correspond to players who were active in the Top 5 leagues but did not transfer during the relevant window and are not included in the regression analysis.

From the 1,066 observations with a transfer fee, a sequence of filters was applied to construct the final analytical sample. First, as mentioned previously, players with fewer than 900 minutes of playing time in the receiving season were removed, as their per-90-minute performance statistics are too volatile to be representative of their true level. Second, goalkeepers were removed on the grounds that their performance metrics are structurally incomparable to those of outfield players. Third, the 178 observations corresponding to the summer 2020 transfer window, identified in the dataset as season year 2021 under the FBref labelling convention, were excluded to avoid the distorting effects of the COVID-19 pandemic on transfer market conditions, as discussed in Section 3.1. These three filters reduced the sample from 1,066 to 795 observations.

Player positions in FBref are reported as compound strings reflecting a player's ability to operate in multiple roles. For example, a player listed as "FW,MF" is primarily a forward but can play as a midfielder. To produce a single positional classification for each player, a hierarchical assignment rule was applied: players with "FW" anywhere in their position string were classified as forwards; remaining players with "MF" were classified as midfielders; all others were classified as defenders. This produced three mutually exclusive positional groups: 302 forwards (38.0%), 275 defenders (34.6%), and 218 midfielders (27.4%). Two binary dummy variables were created, `pos_FW` and `pos_MF`, with defenders serving as the omitted reference category in all regression specifications.

Injury data was joined to the analytical dataset using each player's Transfermarkt profile URL as the join key. Injury records were aggregated to the player-season level by summing all recorded injury days within the relevant season window, producing a single continuous `injury_days` variable per observation. Of the 795 players in the final sample, 490 (61.6%) had at least one recorded injury day in the relevant season, with a mean of 39.3 days among all players and a maximum of 598 days. The remaining 305 players (38.4%) had no recorded injury days and were assigned a value of zero, consistent with the interpretation that the absence of an injury record on Transfermarkt indicates an injury-free period.

It should be noted that six observations in the final sample of 795 player-seasons have missing Transfermarkt market value records. These missing values are due to player profile pages on Transfermarkt that did not return a market value at the relevant season cut-off, typically because the player had only just transferred into a Top 5 league or because Transfermarkt had not yet published a valuation. As market value is a key variable in this study, these six observations are dropped from all regression models in Sets A and B, yielding an effective regression sample of 789 observations. Descriptive statistics in Section 4.1 are reported on the full n = 795 sample for variables that do not depend on market value, and on n = 789 for market value itself. Of the six dropped observations, four are defenders, one a midfielder, and one a forward, which explains the slight differences between the positional counts reported in this section (n = 795) and those used in the regression analysis (n = 789).

The final dataset was exported as both .rds and .csv files for use in the analysis scripts described in Sections 3.5 and 3.6. A complete summary of the variable construction procedures is provided in Section 3.4.

3.4 Measures and scales

This section defines each variable included in the regression models, explains the rationale behind the model, and finally specifies the expected direction of effects based on the theoretical and empirical literature reviewed in Chapter 2. A summary of all variables, their sources, and their hypothesised signs is presented in Table 2.

Table 2. Variable definitions, sources, and expected signs.

Variable	Type	Measurement	Source	Expected Sign
log(transfer_fee)	Dependent (Set B)	Natural log of transfer fee (€)	Transfermarkt	-
log(market_value)	Dependent (Set A) / Independent (Set B)	Natural log of Transfermarkt market value (€)	Transfermarkt	+
Fee-MV Gap	Dependent (Set B)	log(fee) – log(market value)	Computed	-
PC1	Independent	1st principal component (attacking output)	FBref	+
PC2	Independent	2nd principal component (creative/progressive)	FBref	?
PC3	Independent	3rd principal component (residual)	FBref	?

Age	Independent	Age in years	FBref	- (MV), + (Gap)
pos_MF	Control	1 = midfielder (DF = reference)	FBref	?
pos_FW	Control	1 = forward (DF = reference)	FBref	+
injury_days	Independent	Days missed due to injury	Transfermarkt	-
season_year	Control	Season dummies (2023, 2024, 2022 ref)	FBref	?
league	Control	League dummies (PL, Serie A, etc.; Bundesliga ref)	Transfermarkt	±

Note: DF (defender) serves as the reference category for position dummies; Bundesliga and season 2022 serve as reference categories for league and season fixed effects respectively. Source: Author's own compilation.

The dependent variable in all regression models is the natural logarithm of the transfer fee paid, expressed in euros. The raw transfer fee distribution is strongly right-skewed, with a skewness coefficient of 2.85, reflecting the concentration of the transfer market around modest fees with a long upper tail of high-value transactions. The logarithmic transformation reduces this skewness to approximately −0.42 and gives out a distribution that more closely satisfies the normality assumption of OLS regression. Log-transforming the dependent variable also means that regression coefficients are interpreted as the approximate percentage change in transfer fee associated with a one-unit change in the independent variable. The use of a log-transformed fee as the dependent variable is standard practice across the empirical literature and is adopted by all major studies reviewed in Chapter 2 (Dobson & Gerrard, 1999; Müller et al., 2017; Poli et al., 2022, 2024; McHale & Holmes, 2023).

As discussed in Section 3.5, performance variables are not entered directly into the regression but are first reduced to three orthogonal principal components via PCA. The six input variables are: goals per 90 minutes, assists per 90 minutes, expected goals per 90 minutes (xG), non-penalty expected goals per 90 minutes (npxG), progressive carries per 90 minutes, and progressive passes per 90 minutes. All six variables were converted to per-90-minute rates prior to PCA by dividing each raw seasonal count by the number of 90-minute equivalents played, ensuring that players with different amounts of playing time are compared on a common scale. Expected goals is a measure of shot quality that estimates the probability that a given shot results in a goal based on historical data on similar shots; it is a more stable and less luck-dependent measure of attacking contribution than goals scored, and is now being widely used across the football industries (McHale & Holmes, 2023). Progressive carries and progressive passes measure a player's contribution to advancing the ball towards the opponent's goal and

capture dimensions of performance, ball progression, creativity, and technical involvement, that goals and assists alone cannot reflect. The three principal components extracted from these six variables are described in detail in Section 3.5; for the purposes of this section, PC1 is expected to have a positive coefficient reflecting the positive association between attacking output and transfer fees, while PC2 and PC3 carry less determinate prior expectations given their mixed positional interpretation.

Transfermarkt market value is entered as the natural log of the market value in euros, reflecting the same skewness justification as the dependent variable. Market value is the most powerful individual predictor of transfer fees in the empirical literature, with correlations exceeding 0.90 in multiple studies (Herm et al., 2014; Kirschstein & Liebscher, 2019), and its inclusion in the regression is expected to produce a large and highly significant positive coefficient. As discussed in Section 2.4, market value serves as a composite signal that aggregates prior performance history, age trajectory, positional scarcity, and club context into a single crowd-sourced estimate, which explains why it dominates all other predictors when included. Six observations (0.8%) have missing market values and are dropped from models that include this variable. The bivariate correlation between log(market value) and log(transfer fee) in the present sample is 0.780, confirming its central role in the model.

Player age is measured as the integer number of years at the time of the relevant season, as reported by FBref. The theoretical expectation from the literature is an inverted-U relationship, value rising during the early career and declining after the mid-twenties, but in models that control for market value, age is expected to carry a positive coefficient because market value already absorbs the downward aging effect and the remaining variation in age within the transferred sample reflects experience and peak career status (Müller et al., 2017; Franceschi et al., 2024). The mean age in the sample is 24.8 years, with a standard deviation of 3.4 years.

Two binary dummy variables identify midfielders (`pos_MF = 1`) and forwards (`pos_FW = 1`), with defenders serving as the omitted reference category. The choice of defenders as the reference reflects their status as the most homogeneous positional group in terms of transfer fee levels, and ensures that the coefficients on `pos_MF` and `pos_FW` are interpreted relative to a stable baseline. Based on the literature, forwards are expected to command a significant premium relative to defenders (Müller et al., 2017; Majewski, 2016), while the sign for midfielders is less determinate given the heterogeneity of roles within that category.

The injury variable is a perpetual measure of the total number of days missed due to injury in the relevant season, constructed from Transfermarkt injury records as described in Section 3.3. The theoretical expectation is a negative coefficient: greater injury absence signals physical fragility and reduces the number of available match days in the buying club's investment horizon, reducing revenue, sporting contribution, etc. However, as noted in the literature review, injury history is rarely included as an explicit regressor in transfer fee models, making the direction of the empirical finding in this study a substantive contribution. The distribution of injury days is strongly right-skewed, with a mean of 39.3 days and a maximum of 598 days; 305 players (38.4%) have zero recorded injury days in the relevant season.

Season fixed effects are included in all models as binary dummies for season years 2023 and 2024, with season year 2022 (corresponding to the summer 2021 window) serving as the reference category. These dummies control for aggregate inter-annual variation in transfer market activity, including the ongoing post-COVID recovery in 2021, the fee inflation of 2022, and the record-breaking 2023 summer window. League fixed effects, entered as four binary dummies with the Bundesliga as the reference category, are included in a dedicated model specification to test for systematic inter-league differences in fee levels after controlling for player characteristics. The Premier League premium documented by Bell et al. (2024) is expected to produce a large positive coefficient on the Premier League dummy.

3.5 Principal component analysis

The six performance variables described in Section 3.4, goals, assists, expected goals, non-penalty expected goals, progressive carries, and progressive passes, all expressed as per-90-minute rates, show substantial intercorrelation. Entering these variables individually into a regression model would introduce multicollinearity, inflating standard errors, destabilising coefficient estimates, and making individual variable interpretation unreliable. In line with the methodological precedent established by Kirschstein & Liebscher (2019), who applied Principal Component Analysis to address an analogous multicollinearity problem in Bundesliga market value estimation, Principal Component Analysis (PCA) is applied to reduce the six correlated performance variables to a smaller set of orthogonal components that can be entered safely into the regression.

PCA is a dimensionality reduction technique that transforms a set of correlated variables into a new set of uncorrelated variables, the principal components, which are linear combinations of the original variables ordered by the share of total variance they explain (Kirschstein & Liebscher, 2019). Each component is orthogonal to all others by construction,

which eliminates multicollinearity among the performance inputs entirely. The first principal component captures the largest possible share of variance in the original data, the second captures the largest share of remaining variance, and so on. The loadings, the coefficients that define each component as a weighted sum of the original variables, indicate which original variables drive each component and therefore provide the basis for substantive interpretation.

Prior to PCA, all six performance variables were standardised to zero mean and unit variance by computing z-scores. Standardisation is essential before PCA when variables are measured in different units or have substantially different scales, as is the case here: progressive passes per 90 minutes has a mean of approximately 3.49 while goals per 90 minutes has a mean of 0.15, and without standardisation the PCA solution would be dominated by variables with larger absolute values rather than by variables who carry the most information. The PCA was executed in R using the `prcomp()` function with `center = TRUE` and `scale. = TRUE`, which applies the standardisation internally. The analysis was performed on 794 observations, with one observation dropped due to a missing value in a performance variable.

The PCA yielded six components, of which three were retained for use in the regression models. The decision on how many components to retain was guided by two criteria. The Kaiser criterion, retaining components with an eigenvalue greater than 1.0, meaning they explain more variance than a single standardised original variable, suggests retaining two components: PC1 (eigenvalue = 3.21) and PC2 (eigenvalue = 1.45). PC3 has an eigenvalue of 0.66, which falls below the Kaiser threshold. However, the cumulative variance criterion provides a complementary perspective: PC1 and PC2 together explain 77.7% of total performance variance, while adding PC3 brings cumulative explained variance to 88.7%. Given that PC3 still captures approximately 11% of performance variation and that its inclusion adds only one additional regressor to models with 795 observations, the three components were retained. Table 4 presents the variance explained by each component.

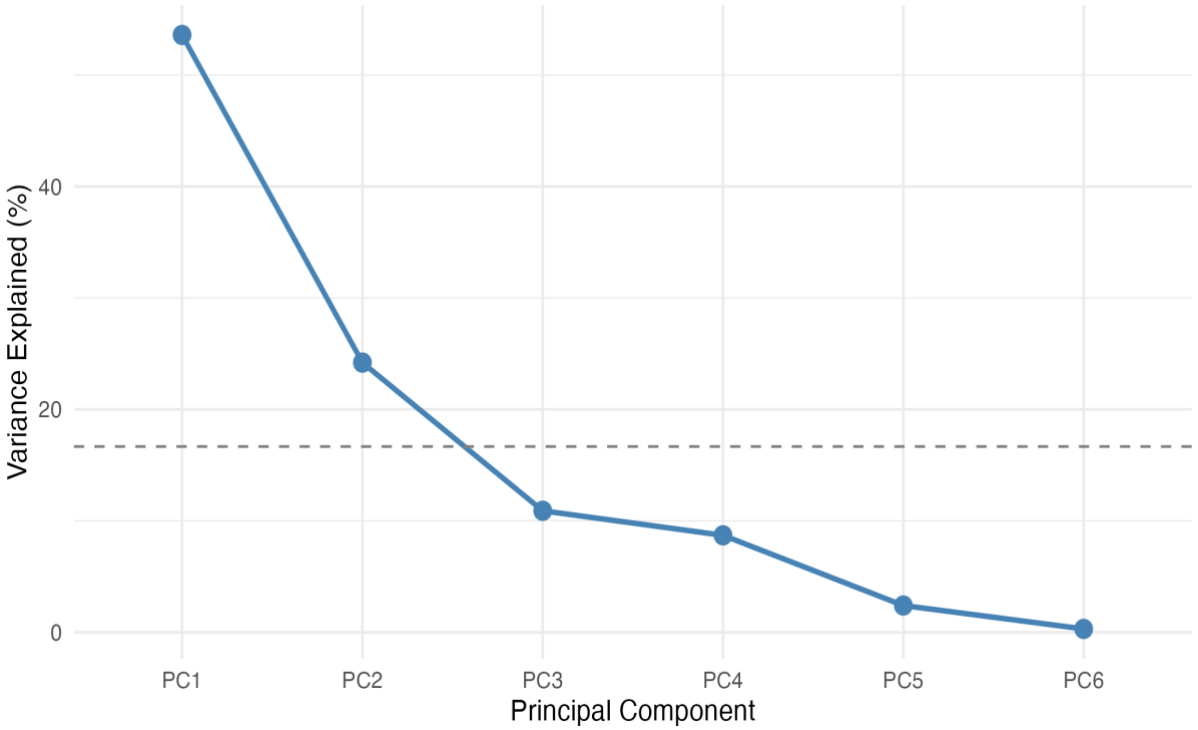


Figure 1. Scree plot of principal components. The dashed line indicates the average variance threshold (16.7%). Source: Author's own calculations. Figure produced using Ggplot2 (Wickham, 2016).

Table 3. Variance explained by principal components.

Component	Variance Explained	Cumulative Variance
PC1	**53.6%**	**53.6%**
PC2	**24.2%**	**77.7%**
PC3	**10.9%**	**88.7%**
PC4	8.7%	97.4%
PC5	2.4%	99.7%
PC6	0.3%	100.0%

Note: Components retained for regression analysis are shown in bold (PC1-PC3). Source: Author's own calculations.

The loadings of the three retained components provide substantive interpretations. PC1, which explains 53.6% of total performance variance, loads most heavily on goals per 90 (0.514), expected goals per 90 (0.541), and non-penalty expected goals per 90 (0.539), with moderate positive loading on assists per 90 (0.278) and progressive carries per 90 (0.177), and a small negative loading on progressive passes per 90 (−0.210). This component therefore reflects a player's attacking output. Players with high PC1 scores are those who score frequently and generate high-quality shots regardless of position; players with low PC1 scores contribute minimally to direct attacking output. PC1 is labelled Attacking Output throughout the analysis.

PC2, explaining 24.2% of total performance variance, loads negatively on progressive carries per 90 (−0.639), progressive passes per 90 (−0.548), and assists per 90 (−0.517), with small positive loadings on goals per 90 (0.077), expected goals per 90 (0.095), and non-penalty expected goals per 90 (0.094). The dominant negative loadings on the three progression and creativity variables indicate that PC2 captures the inverse of creative and technical involvement: a high PC2 score corresponds to low ball-progression and low playmaking output, typical of pure finishers, while a low (negative) PC2 score corresponds to technically proficient playmakers, central midfielders, or wide forwards with high involvement in possession phases. PC2 is labelled Creative and Progressive Involvement throughout the analysis, with the understanding that its regression coefficient is expected to be negative if creative involvement is rewarded by the market.

PC3, explaining a further 10.9% of variance, is dominated by a strong positive loading on progressive passes per 90 (0.799), with moderate negative loadings on assists per 90 (−0.373) and progressive carries per 90 (−0.300). This pattern is less immediately interpretable than the first two components and may reflect a contrast between players who advance the ball through passing versus those who do so through carrying. PC3 is retained for completeness but is not expected to carry a theoretically determinate sign in the regression. The loadings for all three components are presented in Table 5.

Table 4. PCA loadings for the three retained principal components.

Variable	PC1	PC2	PC3
Goals per 90	0.515	0.077	-0.269
Assists per 90	0.277	-0.517	0.318
xG per 90	0.540	0.095	-0.187
npxG per 90	0.539	0.094	-0.152
Progressive carries per 90	0.171	-0.639	0.334
Progressive passes per 90	-0.218	-0.548	-0.810

Note: Loadings represent the correlation between each variable and the principal component. Source: Author's own calculations.

The three principal component scores for each player, PC1, PC2, and PC3, were extracted from the PCA solution and appended to the analytical dataset as continuous variables. These scores are zero-centred by construction and their units are standard deviations of the original standardised performance space. They are fully orthogonal to one another, which is confirmed by the correlation matrix which shows correlations of effectively zero between all three

components. The components were subsequently used as independent variables in all regression models described in Section 3.6.

3.6 Regression specifications

The analytical strategy is organised around two complementary research questions. The first is to determine market value itself: which player characteristics are systematically priced into Transfermarkt's crowd-sourced valuations. The second is to observe and analyse the deviation between market value and realised transfer fees: which factors explain when and why actual negotiated fees exceed or fall short of the crowd's prior assessment. The two questions require two distinct dependent variables and two parallel sets of regression models, estimated on the same sample and with the same explanatory variables to allow direct comparison. All models are estimated via Ordinary Least Squares using the lm() function in R, and all include season fixed effects throughout.

The first set of models takes the natural logarithm of Transfermarkt market value as the dependent variable and estimates a progressive sequence of specifications to identify which characteristics are reflected in the crowd's valuation of a player. Four pooled models are estimated, each nested within the next, alongside three position-specific subgroup regressions and one trimmed-sample robustness check.

Model A1 regresses log(market value) on the three principal components of performance, age, and season fixed effects:

$$\ln(\mathrm{MV})_i = \alpha + \beta_1 \mathrm{PC1}_i + \beta_2 \mathrm{PC2}_i + \beta_3 \mathrm{PC3}_i + \beta_4 \mathrm{Age}_i + \gamma \mathrm{D}_t + \varepsilon_i$$

Model A2 adds position dummies (midfielder and forward, with defender as the reference category) to identify whether positional effects are systematically priced into market value.

Model A3 adds injury days in the relevant season, testing whether the Transfermarkt community penalises injury history when forming valuation estimates.

Model A4 augments Model A3 with four league fixed effects (La Liga, Ligue 1, Premier League, and Serie A, with the Bundesliga as the reference category). This specification tests for systematic inter-league differences in market value that persist after controlling for observable player characteristics, a dimension particularly relevant given the commercial prominence of the Premier League.

$$\ln(\mathrm{MV})_i = \alpha + \beta_1 \mathrm{PC1}_i + \beta_2 \mathrm{PC2}_i + \beta_3 \mathrm{PC3}_i + \beta_4 \mathrm{Age}_i + \beta_5 \mathrm{MF}_i + \beta_6 \mathrm{FW}_i + \beta_7 \mathrm{Injury}_i + \gamma \mathrm{D}_t + \delta \mathrm{L}_k + \varepsilon_i$$

Three position-specific subgroup regressions are then estimated on the defender (n = 271), midfielder (n = 217), and forward (n = 301) subsamples respectively. Each includes PC1, PC2, PC3, age, injury days, and season fixed effects. These models allow the determinants of market value to vary freely across positional groups.

A final robustness check re-estimates Model A4 on a trimmed sample from which observations below the 5th percentile and above the 95th percentile of market value have been removed, testing whether the core findings are robust to extreme valuations.

The second set of models investigates which factors explain systematic deviations between realised transfer fees and the market value baseline. The dependent variable is constructed as the difference between the log-transformed transfer fee and the log-transformed market value:

$$\mathrm{Gap}_i = \ln(\mathrm{Fee})_i - \ln(\mathrm{MV})_i$$

This formulation is equivalent to the log of the fee-to-market-value ratio, and a positive value indicates that the realised fee exceeds the crowd's prior valuation while a negative value indicates the opposite. Using this transformation rather than the ratio itself keeps the interpretability of coefficients as approximate percentage deviations. It also removes market value from the right-hand side of the regression.

The same progressive structure is applied. Model B1 regresses the gap on performance components, age, and season fixed effects:

$$\mathrm{Gap}_i = \alpha + \beta_1 \mathrm{PC1}_i + \beta_2 \mathrm{PC2}_i + \beta_3 \mathrm{PC3}_i + \beta_4 \mathrm{Age}_i + \gamma \mathrm{D}_t + \varepsilon_i$$

Model B2 adds position dummies. Model B3 adds injury days. Model B4 adds league fixed effects to produce the full pooled specification:

$$\mathrm{Gap}_i = \alpha + \beta_1 \mathrm{PC1}_i + \beta_2 \mathrm{PC2}_i + \beta_3 \mathrm{PC3}_i + \beta_4 \mathrm{Age}_i + \beta_5 \mathrm{MF}_i + \beta_6 \mathrm{FW}_i + \beta_7 \mathrm{Injury}_i + \gamma \mathrm{D}_t + \delta \mathrm{L}_k + \varepsilon_i$$

A trimmed-sample robustness check re-estimates Model B4 on observations between the 5th and 95th percentiles of transfer fee. Position-specific subgroup regressions are estimated but not reported in the main text, as the limited within-group variation in the fee-MV gap produces weak and largely non-significant results.

Variance Inflation Factors were computed for the full models (A4 and B4) to verify that the remaining regressors, age, position, injury days, league, do not introduce problematic multicollinearity. All VIF values are well below the conventional threshold of 5.

Together, the two sets of models provide a two-stage account of the price formation process in the European football transfer market. Set A identifies which player characteristics the Transfermarkt community systematically prices into its market valuations, while Set B identifies which characteristics explain the systematic component of the deviation between those crowd valuations and the fees actually paid by buying clubs. Goodness-of-fit statistics for all specifications are summarised in Table 6.

Table 5. Summary of regression specifications and goodness-of-fit statistics.

Model	Specification	n	R^2	Adj. R^2
A1	log(MV): PC1-3, Age, season FE	789	0.339	0.334
A2	A1 + position dummies	789	0.363	0.356
A3	A2 + injury days	789	0.363	0.355
A4	A3 + league fixed effects	789	0.514	0.506
A_DF	Defenders only	271	0.229	0.209
A_MF	Midfielders only	217	0.353	0.332
A_FW	Forwards only	301	0.515	0.503
B1	Gap: PC1-3, Age, season FE	789	0.031	0.023
B2	B1 + position dummies	789	0.043	0.033
B3	B2 + injury days	789	0.061	0.051
B4	B3 + league fixed effects	789	0.103	0.088

Note: Set A models the determinants of market value (log of market value as dependent variable); Set B models the fee–market value gap (log(fee) − log(MV)). FE = fixed effects. Source: Author's own calculations.

4. Results

This chapter presents the empirical findings of the study in three sections. Section 4.1 provides descriptive statistics for the key variables and examines the bivariate relationships among them. Section 4.2 presents the results of the two parallel sets of progressive pooled regression models. Section 4.3 discusses the position-specific subgroup regressions and the outlier-trimming robustness checks.

4.1 Descriptive statistics

Table 6 presents summary statistics for the key variables included in the regression analysis across the 795 player-season observations in the final sample.

Table 6. Descriptive statistics for key variables (n = 795).

Variable	Mean	SD	Min	Max
Transfer fee (€)	14,300,579	18,360,483	70,000	127,000,000
Market value (€)	17,210,773	19,879,355	600,000	180,000,000
Age	24.8	3.44	17	37
Injury days	39.29	66.60	0	598
Minutes per 90	20.27	6.92	10	38
PC1 (Attacking Output)	0.00	1.81	-2.29	8.64
PC2 (Creative Involvement)	0.00	1.22	-7.32	2.28
PC3 (Residual)	0.00	0.81	-4.53	4.05

Note: Transfer fee and market value expressed in euros. Market value reported on 789 observations due to 6 missing values. Source: Author's own calculations.

The distribution of transfer fees is strongly right-skewed, with a mean of approximately €14.3 million and a median of €8.0 million, a gap of €6.3 million that reflects the concentration of the market around modest fees and a relatively small number of very high-value transactions. The maximum observed fee of €127 million corresponds to Jude Bellingham's transfer from Borussia Dortmund to Real Madrid in the summer of 2023, while the minimum of €70,000

represents a small permanent transaction at the lower end of the Top 5 league market. The standard deviation of €18.4 million largely exceeds the mean, confirming the high dispersion of the fee distribution and reinforcing the necessity of the logarithmic transformation applied to the dependent variable throughout the analysis. The log-transformed fee distribution has a mean of 15.81 and a standard deviation of 1.25, indicating a substantially more symmetric distribution suitable for OLS regression.

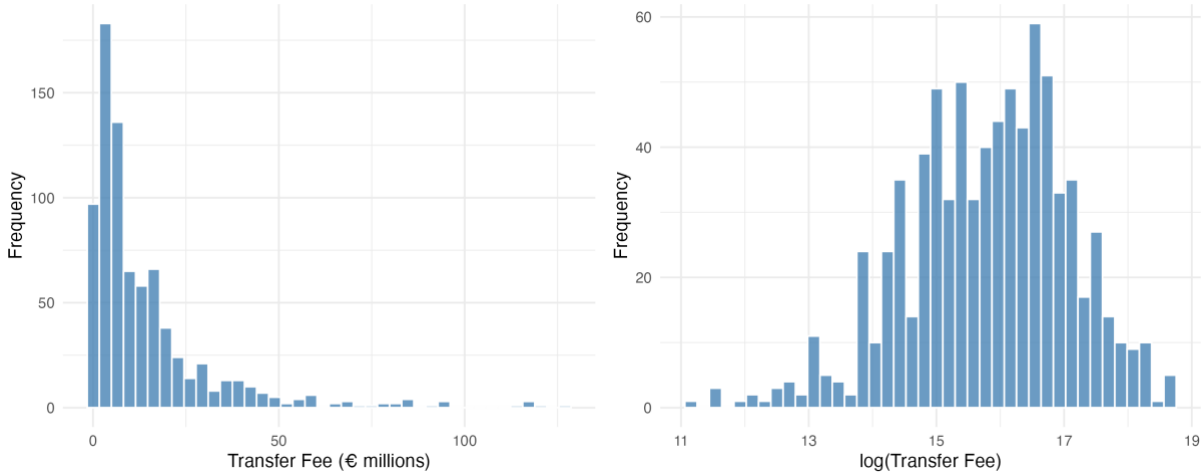


Figure 2. Distribution of transfer fees before and after log transformation (n = 795). Source: Author's own calculations. Figure produced using Ggplot2 (Wickham, 2016).

Transfermarkt market values display a similar distributional pattern, with a mean of €17.2 million and a median of €10.0 million. The mean market value exceeds the mean transfer fee by approximately €2.9 million, consistent with Poli's et al. (2022) observation that Transfermarkt values tend to slightly overestimate actual transaction prices on average. The mean player age in the sample is 24.8 years, with a standard deviation of 3.4 years and a range of 17 to 37, confirming that the transferred player population is concentrated in the early-to-mid career phase consistent with the literature's inverted-U age-value relationship. The mean injury burden of 39.3 days, with a standard deviation of 66.6 days and a maximum of 598 days, reflects the highly skewed nature of injury incidence: the majority of players experience either no injuries or short absences, while a small number of players missed extensive periods. Playing time, measured in per-90-minute equivalents, ranges from the minimum threshold of 10 (corresponding to exactly 900 minutes) to a maximum of 38, with a mean of 20.3, indicating that the sample is not dominated by fringe players but represents a broad cross-section of regularly used squad members.

The principal component scores are zero-centred by construction, with standard deviations of 1.81, 1.22, and 0.81 for PC1, PC2, and PC3 respectively.

Table 7 presents the bivariate correlation matrix among the key continuous variables.

Table 7. Bivariate correlation matrix for key variables.

	log(fee)	PC1	PC2	PC3	Age	log(MV)	Injury days
log(fee)	1.000						
PC1	0.260	1.000					
PC2	-0.149	-0.015	1.000				
PC3	-0.155	0.006	0.003	1.000			
Age	-0.241	0.013	0.062	-0.143	1.000		
log(MV)	0.780	0.283	-0.230	-0.175	-0.401	1.000	
Injury days	0.095	0.064	-0.006	0.023	0.062	0.004	1.000

Note: log(fee) = natural log of transfer fee; log(MV) = natural log of Transfermarkt market value. Triangular display: only the lower triangle is reported as the matrix is symmetric. Source: Author's own calculations.

The most striking feature of the correlation matrix is the strong positive correlation between log(market value) and log(transfer fee) of 0.780, which immediately shows the close relationship between the crowd-sourced market valuation and the realised transfer fee. PC1, the attacking output component, exhibits the strongest performance-fee correlation at 0.260, while PC2 and PC3 are more weakly correlated at −0.149 and −0.155 respectively. Age is negatively correlated with log(transfer fee) at −0.241, consistent with the finding that younger players command higher fees. Injury days display only a very weak positive raw correlation with log(transfer fee) of 0.095, suggesting that the injury history of a player is not a strong standalone predictor of fees.

The correlation structure among the independent variables reveals two main patterns. First, log(market value) is negatively correlated with age at −0.401, confirming that Transfermarkt systematically assigns higher values to younger players. Second, PC1 (attacking output) is positively correlated with log(market value) at 0.283, while PC2 (creative and progressive involvement) is negatively correlated with log(market value) at −0.230, indicating that Transfermarkt valuations reward goal-scoring metrics more heavily than ball-progression metrics.

4.2 Regression results

This section presents the results of the two sets of progressive OLS regression models described in Section 3.6. First, Set A, which identifies the determinants of Transfermarkt market

value, is discussed; second, Set B, which explains the systematic deviations between realised transfer fees and market value, is presented. Tables 8 and 9 present the full results for Sets A and B respectively.

Table 8. Determinants of Market Value (Set A).

	Dependent variable:			
	log(Market Value)			
	(1)	(2)	(3)	(4)
PC1 (Attacking Output)	0.172****	0.253****	0.253****	0.249****
	(0.017)	(0.024)	(0.024)	(0.021)
PC2 (Creative Involvement)	-0.178****	-0.197****	-0.197****	-0.201****
	(0.026)	(0.026)	(0.026)	(0.023)
PC3 (Residual)	-0.318****	-0.311****	-0.311****	-0.319****
	(0.039)	(0.041)	(0.041)	(0.037)
Age	-0.135****	-0.135****	-0.135****	-0.130****
	(0.009)	(0.009)	(0.009)	(0.008)
Midfielder (ref: Defender)		-0.221***	-0.218**	-0.208***
		(0.085)	(0.086)	(0.075)
Forward (ref: Defender)		-0.523****	-0.522****	-0.511****
		(0.098)	(0.098)	(0.086)
Injury Days			0.0001	-0.0003
			(0.0005)	(0.0004)
Season 2023	0.023	0.010	0.010	-0.004
	(0.081)	(0.080)	(0.080)	(0.070)
Season 2024	0.022	0.034	0.035	0.076
	(0.080)	(0.079)	(0.079)	(0.069)
La Liga				-0.053
				(0.099)
Ligue 1				-0.156*
				(0.090)
Premier League				0.947****
				(0.087)
Serie A				0.005
				(0.087)
Constant	19.431****	19.694****	19.692****	19.382****
	(0.237)	(0.239)	(0.239)	(0.218)
Observations	789	789	789	789
R^2	0.339	0.363	0.363	0.514
Adjusted R^2	0.334	0.356	0.355	0.506

*Note: Standard errors in parentheses. Reference categories: Defenders (position), Bundesliga (league), 2022 (season). Significance levels: *p<0.1; **p<0.05; ***p<0.01; ****p<0.001. Source: Author's own calculations. Tables generated using Stargazer (Hlavac, 2022).*

The baseline model regressing log(market value) on the three principal components, age, and season fixed effects achieves an R^2 of 0.339 (Adjusted $R^2 = 0.334$, $F = 66.94$, $p < 0.001$). Player performance and age alone account for approximately one-third of the variation in Transfermarkt market valuations. All three principal components are highly significant. PC1, the Attacking Output component, carries a positive coefficient of 0.172 ($p < 0.001$), indicating that a one-standard-deviation increase in attacking output is associated with an approximately 17.2% increase in market value. PC2, the Creative and Progressive Involvement component, carries a negative coefficient of −0.178 ($p < 0.001$), while PC3 is also negative and highly significant ($\beta = -0.318$, $p < 0.001$). Age carries a strong negative coefficient of −0.135 ($p < 0.001$), meaning that each additional year of age is associated with approximately a 13.5% reduction in market value, confirming the youth premium predicted by Rosen (1981) and explained across the literature (Müller et al., 2017; Franceschi et al., 2024). Season fixed effects are statistically non-significant.

The inclusion of position dummies in Model A2 raises R^2 modestly to 0.363 (Adjusted $R^2 = 0.356$) and reveal large positional effects. Midfielders have market values approximately 22% lower than equivalently measured defenders ($\beta = -0.221$, $p = 0.010$), and forwards have values approximately 52% lower ($\beta = -0.523$, $p < 0.001$). This result may appear counterintuitive since forwards seem the most present in the commercial and media landscape of football, but it reveals that the defender population that transfers is systematically more elite than the forward or midfielder populations. Forwards and midfielders transfer at much higher volume and cover a broader quality spectrum, including many modestly valued players, whereas defenders who transfer to Top 5 league clubs tend to be high-profile acquisitions. The coefficients on PC1, PC2, PC3, and age remain stable and highly significant.

The addition of injury days in Model A3 changes essentially nothing in the model fit ($R^2 = 0.363$, Adjusted $R^2 = 0.355$) and the injury coefficient is statistically non-significant ($\beta = 0.00013$, $p = 0.781$). This is a crucial finding in itself: the Transfermarkt community does not systematically penalise injury history when forming market valuations, at least in the aggregate. Players who have missed more days of play in the relevant season are valued no lower than those who have missed fewer, conditional on their performance output, age, and position.

The addition of league fixed effects in Model A4 produces the most substantial improvement in fit of any single variable block, with R^2 rising from 0.363 to 0.514 (Adjusted

$R^2 = 0.506$, $F = 63.17$, $p < 0.001$), an increase of 15.2 percentage points. The Premier League coefficient is both large and highly significant ($\beta = 0.947$, $p < 0.001$), indicating that Premier League players have market values approximately 158% higher than Bundesliga players with similar characteristics (exp(0.947) = 2.58). La Liga, Serie A, and Ligue 1 do not differ significantly from the Bundesliga baseline, with Ligue 1 in fact showing a weak negative coefficient ($\beta = -0.156$, $p = 0.083$). The dominance of the Premier League premium over other league effects confirms the findings of Bell et al. (2024) by demonstrating that the Premier League commercial advantage is priced directly into crowd-sourced market valuations, not just in the actual realised transfer fees. The PC1, PC2, PC3, age, and position coefficients all remain highly significant and close in magnitude to Model A3.

Table 9. Determinants of the Fee-Market Value Gap (Set B).

	Dependent variable:			
	log(Fee) - log(Market Value)			
	(1)	(2)	(3)	(4)
PC1 (Attacking Output)	0.013	-0.033	-0.038*	-0.032
	(0.016)	(0.022)	(0.021)	(0.021)
PC2 (Creative Involvement)	0.043*	0.048**	0.051**	0.049**
	(0.023)	(0.024)	(0.024)	(0.023)
PC3 (Residual)	0.012	-0.007	-0.007	-0.010
	(0.035)	(0.037)	(0.037)	(0.037)
Age	0.037****	0.037****	0.034****	0.036****
	(0.008)	(0.008)	(0.008)	(0.008)
Midfielder (ref: Defender)		0.022	0.054	0.068
		(0.077)	(0.077)	(0.075)
Forward (ref: Defender)		0.259***	0.276***	0.272***
		(0.089)	(0.088)	(0.086)
Injury Days			0.002****	0.002****
			(0.0004)	(0.0004)
Season 2023	-0.036	-0.031	-0.034	-0.038
	(0.072)	(0.072)	(0.071)	(0.070)
Season 2024	-0.027	-0.031	-0.020	-0.009
	(0.071)	(0.071)	(0.071)	(0.069)
La Liga				0.136
				(0.099)
Ligue 1				0.279***
				(0.090)
Premier League				0.499****
				(0.088)

Serie A				0.302****
				(0.087)
Constant	-1.210****	-1.312****	-1.338****	-1.651****
	(0.212)	(0.216)	(0.214)	(0.219)
Observations	789	789	789	789
R^2	0.031	0.043	0.061	0.103
Adjusted R^2	0.023	0.033	0.051	0.088

*Note: Standard errors in parentheses. Reference categories: Defenders (position), Bundesliga (league), 2022 (season). Significance levels: *p<0.1; **p<0.05; ***p<0.01; ****p<0.001. Source: Author's own calculations. Tables generated using Stargazer (Hlavac, 2022).*

Set B takes the logarithmic gap between transfer fee and market value as the dependent variable, isolating the systematic component of deviation from the crowd's prior valuation. Overall explanatory power in these models is substantially lower than in Set A, with R^2 values ranging from 0.03 to 0.10, indicating that most variation in the fee-MV gap is idiosyncratic negotiation noise. The 10% of variance that is systematically explained still nonetheless reveals patterns about when, why and how fees differ from market values.

In the baseline specification regressing the gap on principal components, age, and season fixed effects, only age carries a statistically significant coefficient ($\beta = 0.037$, $p < 0.001$). Each additional year of age is associated with a 3.7% increase in the fee-market value gap. Conditional on market value, older players command fees that exceed their crowd valuations by more than younger players do, confirming the interpretation that Transfermarkt discounts age more heavily in market valuations than actual clubs do when negotiating transfers.

Model B2 reveals the strongest substantive finding of the Set B analysis: forwards command fees approximately 26% higher than their market values would predict ($\beta = 0.259$, $p = 0.004$). The market systematically overpays for forwards relative to Transfermarkt's crowd assessment, consistent with the positional scarcity arguments of Rosen (1981) and Majewski (2016). The midfielder dummy is not statistically distinguishable from the defender baseline ($\beta = 0.022$, $p = 0.775$). PC2 becomes significant at the 5% level ($\beta = 0.048$, $p = 0.044$), suggesting a slight premium for creative and progressive involvement above the market value baseline. The age premium persists at 3.7% per year.

Injury days enters the gap regression as positive and highly significant ($\beta = 0.0017$, $p < 0.001$). This is similar to the finding from the original analytical framework and retains the same substantive interpretation: the fee-MV gap is systematically wider for players with more recorded injury days, reflecting a selection effect in which the players with the most game time and media coverage accumulate both more extensive injury records and higher fee-MV premia

because their transactions are the most extensively negotiated ones. The forward premium remains significant ($\beta = 0.276$, $p = 0.002$) and the age effect is stable.

The full specification with league fixed effects achieves $R^2 = 0.103$ (Adjusted $R^2 = 0.088$), the highest of the Set B pooled models. League effects explain approximately 4 percentage points of the fee-MV gap variation and reveal a coherent pattern. The Premier League commands a gap premium of 0.499 log points ($p < 0.001$), meaning Premier League clubs pay fees approximately 65% above the market value baseline ($\exp(0.499) = 1.65$). Serie A clubs pay a premium of approximately 35% ($\beta = 0.302$, $p < 0.001$), and Ligue 1 clubs approximately 32% ($\beta = 0.279$, $p = 0.002$). La Liga is the only league whose premium is not statistically distinguishable from the Bundesliga baseline ($\beta = 0.136$, $p = 0.172$). Taken together, these coefficients indicate that the buying clubs in Serie A, Ligue 1, and especially the Premier League systematically overpay relative to crowd-sourced valuations, while Bundesliga and La Liga clubs transact closer to market-value benchmarks. The forward premium, age effect, and injury effect all remain significant.

4.3 Robustness and position-specific analyses

This section presents the position-specific regressions for Set A, which examine whether the determinants of market value differ systematically across playing positions, and the two trimmed-sample robustness checks that test the sensitivity of the core findings to extreme observations. Position-specific regressions for Set B were estimated but are not reported in the main text, as they produced weak and largely non-significant results; full outputs are available in the Appendix.

Table 10 presents the three subgroup regressions estimating Model A3 separately on the defender (n = 271), midfielder (n = 217), and forward (n = 301) subsamples. Each model includes PC1, PC2, PC3, age, injury days, and season fixed effects.

Table 10. Position-Specific Determinants of Market Value.

	Dependent variable:		
	log(Market Value)		
	Defenders	Midfielders	Forwards
	(1)	(2)	(3)
PC1 (Attacking Output)	0.132	0.207***	0.301****
	(0.095)	(0.073)	(0.025)
PC2 (Creative Involvement)	-0.143**	-0.114	-0.245****
	(0.065)	(0.075)	(0.033)
PC3 (Residual)	-0.345****	-0.404****	-0.251****

	(0.100)	(0.092)	(0.060)
Age	-0.127****	-0.156****	-0.131****
	(0.019)	(0.018)	(0.013)
Injury Days	0.0004	0.001	-0.001
	(0.001)	(0.001)	(0.001)
Season 2023	-0.064	0.192	-0.007
	(0.147)	(0.159)	(0.115)
Season 2024	-0.100	0.227	0.018
	(0.153)	(0.152)	(0.111)
Constant	19.396****	19.772****	19.017****
	(0.475)	(0.437)	(0.334)
Observations	271	217	301
R^2	0.229	0.353	0.515
Adjusted R^2	0.209	0.332	0.503

*Note: Standard errors in parentheses. Reference category: 2022 (season). Sample sizes: Defenders n=271, Midfielders n=217, Forwards n=301. Significance levels: *p<0.1; **p<0.05; ***p<0.01; ****p<0.001. Source: Author's own calculations. Tables generated using Stargazer (Hlavac, 2022).*

The three position-specific models reveal a theoretically meaningful asymmetry in how performance metrics are priced into market value across positional groups. The forward model achieves an R^2 of 0.515 (Adjusted R^2 = 0.503), much higher than either the midfielder model (R^2 = 0.353) or the defender model (R^2 = 0.229). This gap shows that the performance variables that enter the PCA, goals, assists, expected goals, progressive carries, and progressive passes, capture the core dimensions of forward play almost completely, while they capture only a fraction of what determines defensive or midfield value.

In the forward subgroup, all three principal components and age are highly significant. PC1 carries a coefficient of 0.301 ($p < 0.001$), meaning that a one-standard-deviation increase in attacking output raises a forward's market value by approximately 30%. PC2 carries a significant negative coefficient ($\beta = -0.245$, $p < 0.001$), and PC3 is also significantly negative ($\beta = -0.251$, $p < 0.001$). Age has a pronounced negative effect ($\beta = -0.131$, $p < 0.001$): each additional year reduces a forward's market value by approximately 13%. Injury days and season dummies are not significant.

In the midfielder subgroup, PC1 remains significant and positive ($\beta = 0.207$, $p = 0.005$), but PC2 becomes statistically non-significant ($\beta = -0.114$, $p = 0.127$). PC3 is significant and negative ($\beta = -0.404$, $p < 0.001$), and age remains strongly negative ($\beta = -0.156$, $p < 0.001$), the largest age discount of any positional group. The fact that midfielders' market values penalise age more heavily than forwards' or defenders' reflects that markets take into account

that midfield positions are the most physically intense ones, making midfielders more vulnerable to aging.

In the defender subgroup, the explanatory power of performance metrics collapses. PC1 is not statistically significant ($\beta = 0.132$, $p = 0.163$), and PC2 is significant only at the 5% level ($\beta = -0.143$, $p = 0.028$). PC3 remains significant ($\beta = -0.345$, $p < 0.001$), and age carries a negative coefficient of -0.127 ($p < 0.001$). The low R^2 of the defender model (0.229) reflects a deeper limitation: the six input variables to the PCA are overwhelmingly offensive, and therefore contain almost no information about what makes a defender valuable, aerial ability, positional discipline, tackling, leadership, and defensive reading of the game. The Transfermarkt community's valuation of defenders clearly incorporates these dimensions, but the variables available through FBref cannot capture them. The defender model's R^2 of 22.9% should therefore be interpreted as a lower bound on what performance metrics can explain for this positional group, conditional on the specific variables used.

Injury days are non-significant in all three position-specific models, confirming the earlier finding that the Transfermarkt community does not systematically discount market values for injury history regardless of playing position.

Model A4 was re-estimated on a trimmed sample from which observations below the 5th percentile and above the 95th percentile of market value were removed, leaving 724 observations spanning market values from approximately €2 million to €55 million. The trimmed model achieves $R^2 = 0.424$ (Adjusted $R^2 = 0.414$), a decrease of 9.1 percentage points compared to the full-sample Model A4 that is due to the reduced variance in the dependent variable. All substantive findings are preserved. PC1 remains positive and significant ($\beta = 0.220$, $p < 0.001$), PC2 negative and significant ($\beta = -0.177$, $p < 0.001$), and PC3 negative and significant ($\beta = -0.271$, $p < 0.001$). Age carries a coefficient of -0.107 ($p < 0.001$), slightly smaller in magnitude than in the full sample, indicating that the age discount is somewhat steeper at the extremes of the market value distribution. The forward and midfielder dummies remain significant and negative. The Premier League premium persists at 0.885 log points ($p < 0.001$), confirming that the league effect is not only driven by superstar transactions at the top of the market.

Model B4 was re-estimated on a trimmed sample spanning transfer fees between approximately €1.2 million and €49 million ($n = 709$). The trimmed model achieves $R^2 = 0.120$ (Adjusted $R^2 = 0.104$), marginally higher than the full-sample Model B4. The forward premium remains significant ($\beta = 0.265$, $p < 0.001$), age remains positive and significant ($\beta = 0.044$, $p < 0.001$), and injury days remains positive and significant ($\beta = 0.0012$, $p = 0.001$). The league

effects retain their ordering but at a lower scale: the Premier League premium falls from 0.499 to 0.245 log points ($p = 0.002$), Serie A from 0.302 to 0.203 ($p = 0.007$), and Ligue 1 from 0.279 to 0.169 ($p = 0.029$). La Liga remains non-significant. The fact that league premia attenuate when the fee distribution is trimmed indicates that a meaningful portion of the Premier League premium is driven by the highest-value transactions, the superstar transfers that most visibly deviate from market value estimates. The reduced-magnitude premium of 0.245 log points in the trimmed sample is nevertheless substantial, corresponding to approximately 28% above the Bundesliga baseline after controlling for player characteristics. Notably, PC1 becomes marginally significant and negative in the trimmed Set B model ($\beta = -0.046$, $p = 0.016$), suggesting that within the middle of the fee distribution, forwards with higher attacking output have a slightly smaller fee premium over their market values, a subtle pattern consistent with the market's tendency to fully price in the goal-scoring ability of elite forwards while leaving more room for premium bidding on slightly lower-profile players.

Taken together, the position-specific and trimmed-sample analyses confirm that the core findings of both Set A and Set B are robust. The distinctive pattern of positional asymmetry in market value determination, performance metrics explaining half of forward valuation but less than a quarter of defender valuation, is a substantive contribution of the present study, and one that clarifies why pooled models of player valuation typically underperform relative to position-specific specifications in the literature (Majewski, 2016; Poli et al., 2022).

5. Discussion

This chapter discusses the empirical results presented in Chapter 4 considering the literature reviewed in Chapter 2. Section 5.1 examines the key findings of both sets of regressions and interprets them in relation to the theoretical and empirical literature on football player valuation. Section 5.2 explores the implications of the position-specific analysis and the limits of performance metrics as determinants of player value. Section 5.3 puts the findings in a broader cross-sport comparative framework drawing on the sports economics literature. Section 5.4 acknowledges the limitations of the study.

5.1 Key findings and interpretation

The central questions motivating this study can now be answered directly on the basis of the empirical evidence presented in Chapter 4. The first question, what determines Transfermarkt market value, is answered by Set A. Market value is explained to a substantial degree ($R^2 = 0.515$) by a combination of performance metrics, age, position, and league of destination. Player age is the single most influential continuous predictor, with each additional year associated with an approximately 13.5% reduction in market value. Performance metrics capture a meaningful share of valuation variance, but their effects vary widely across positions. And league of destination, particularly the Premier League, exerts a major influence on market value that is independent of player characteristics, consistent with the commercial and financial premium of the English top league documented across the literature (Frick, 2007; Franceschi et al., 2024).

The second question, which factors explain systematic deviations between realised transfer fees and market value estimates, is answered by Set B. Most variation in the fee-market value gap is idiosyncratic ($R^2 = 0.103$), reflecting the negotiation dynamics, urgency considerations, and club-specific financial circumstances that Carmichael & Thomas (1993) identified as crucial to fee determination. Yet the systematic component of that deviation is structured and interpretable. Three findings stand out: forwards command fees approximately 26% above their market values, Premier League clubs pay fees approximately 65% above the market value of a player, and players with higher injury days accumulate systematically larger fee-MV premium due to a selection effect.

The finding that age carries a strong negative coefficient in Set A ($\beta = -0.135$, $p < 0.001$) but a positive coefficient in Set B ($\beta = +0.037$, $p < 0.001$) reveals an important asymmetry in how age is priced between the crowd-sourced market valuation and the realised transfer market. Transfermarkt systematically discounts age, consistent with the youth premium predicted by

Rosen's (1981) superstar framework and extensively documented in empirical player valuation research (Müller et al., 2017; Franceschi et al., 2024). The buying clubs involved in actual transactions, however, appear to discount age less aggressively: among transferred players, older players command fees that exceed their market values by more than younger players do. One interpretation, consistent with Carmichael & Thomas (1993), is that clubs value the reduced uncertainty associated with proven players, an older player provides more seasonal information than a younger player. The transferred older player therefore commands a premium above the crowd's age-adjusted valuation because the buying club is paying for demonstrated, proven, measurable quality and reduced option-value risk.

Forwards emerge as a particularly distinctive positional group across both sets of models. In Set A, forwards have market values approximately 52% lower than equivalent defenders, showing that the number of transferred forwards varies much more widely in quality than the transferred defender population, with forwards being transferred more often. In Set B, however, forwards command fees that exceed their market values by approximately 26%. The two findings are complementary rather than contradictory: Transfermarkt values transferred forwards on the same scale as defenders on average, while actual buying clubs pay a premium above those conservative valuations when they acquire forwards. This is consistent with the superstar theory of Rosen (1981) and the goal-scoring scarcity premium documented by Majewski (2016): in a low-scoring sport like football, proven forwards command fees that exceed even sophisticated prior estimates of their value, because their skill is in the tail of the distribution and difficult to substitute.

The Premier League premium is the most consistent and economically significant finding of the entire analysis. In Set A, Premier League players have market values approximately 158% higher than equivalent Bundesliga players, with other leagues statistically indistinguishable from the Bundesliga benchmark. In Set B, Premier League clubs pay fees approximately 65% above the market value baseline, placing Serie A and Ligue 1 in the middle and La Liga as the only league not significantly distinguishable from the Bundesliga. These dual findings extend the work of Bell et al. (2024) by demonstrating that the English commercial advantage is priced twice into the transfer market: first into Transfermarkt's market value, and then again into the fees that Premier League buying clubs pay above those valuations. The cumulative effect is that a player moving to the Premier League commands a fee that substantially exceeds what a comparable player moving to the Bundesliga would command.

The injury days variable behaves differently across the two sets. In Set A, injury days are statistically non-significant ($\beta = 0.00013$, $p = 0.781$), indicating that the Transfermarkt

community does not systematically discount market values for injury history. In Set B, injury days are positive and highly significant ($\beta = 0.0017$, $p < 0.001$). Taken together, these findings suggest that the injury effect operates not through the valuation stage but through the transfer selection stage: players who accumulate more injury records are the same players who play the most games (increasing chances of injury exponentially), whose transfers are most extensively negotiated and most thoroughly documented, producing a logical correlation between injury days and the fee-MV gap that reflects player prominence rather than clubs pricing injury risk.

5.2 Position heterogeneity and the limits of performance metrics

The position-specific models in Set A reveal a major asymmetry with direct implications for the quantitative study of football player value. The forward model achieves $R^2 = 0.515$, the midfielder model $R^2 = 0.353$, and the defender model $R^2 = 0.229$. Performance metrics, as operationalised through the three principal components, capture half of forward market value variance but less than a quarter of defender market value variance. This gap thus defines what a detailed performance variable set like FBref can and cannot measure.

The PCA inputs, goals, assists, expected goals, non-penalty expected goals, progressive carries, and progressive passes, are overwhelmingly offensive in character. For forwards, they capture the essential dimensions of the position's value contribution. For midfielders, they capture a substantial share but don't capture crucial parts of the midfield's duty like defensive work rate, positional discipline, and passing range. For defenders, they capture almost nothing of the skills that determine defensive value: aerial ability, tackling, interception reading, and organisational leadership. The low R^2 in the defender model is therefore not a failure of the statistical approach but a limitation of the variable set available through public platforms, and an illustration of the judge principle articulated by Herm et al. (2014). The Transfermarkt community prices defenders on information that researchers working with FBref data alone cannot access.

The forward subgroup also produces the largest Premier League premium in the Set A analysis and the largest forward-specific gap premium in Set B. The combined interpretation is that the forward market is the most superstar-like segment of the transfer market: small differences in attacking output translate into large differences in valuation, the Premier League commands a structural premium for forwards that does not appear for other positions, and realised fees systematically exceed market value estimates for this group. This pattern aligns closely with Lucifora & Simmons's (2003) empirical confirmation of superstar wage effects in Italian football using quantile regression.

5.3 Cross-sport comparison

The two-stage analytical framework adopted in this study, first modelling the determinants of a market-based valuation signal, then modelling systematic deviations between that signal and realised compensation, has direct input in the literature on other professional sports, though with structural differences that illuminate the distinctive features of football's transfer system.

In Major League Baseball, Scully (1974) pioneered the empirical estimation of player marginal revenue products, demonstrating that pitching and hitting statistics are strong predictors of team win percentage and therefore of a player's economic value. Scully's approach was ultimately a one-stage framework: marginal revenue product was computed directly from performance statistics and then compared to observed salaries, with the gap attributed to undervaluation of players due to then active reserve clause system. In the absence of a persistent crowd-sourced valuation mechanism analogous to Transfermarkt, baseball research has not typically adopted the two-stage framework used here. The abolition of the reserve clause and the introduction of free agency in 1976 dramatically increased the correspondence between performance and salary (Kahn, 2000), producing a market more similar to the post-Bosman European football context.

In the National Hockey League, Idson & Kahane (2000) found that individual player characteristics, points scored, plus-minus ratings, physical attributes, and star status, explained a substantial share of salary variance, but that team-level variables contributed additional explanatory power by modifying the returns to individual performance. Defensemen with scoring ability commanded higher salaries than equivalently productive forwards because scoring was scarcer among defensemen. In both cases, positional scarcity modifies the market's pricing of performance: the rarest skill in a position commands the greatest premium, whether that skill is scoring ability in a defensive hockey player or proven goal-scoring in a football forward.

The most general cross-sport observation, noted by Kahn (2000) and applicable to the present findings, is that professional sports labour markets exhibit strong league-specific and structural elements that systematically affect the correspondence between performance and compensation. In football, the Premier League's structural dominance of the transfer market has no direct parallel in North American sports, where league-level compensation differentials are largely absent due to salary caps and centralised labour agreements.

5.4 Limitations

Several limitations of the present study merit acknowledgement. First, the absence of a popularity variable, supported by the theoretical framework of Rosen (1981), Hofmann et al. (2021), and Malagón-Selma et al. (2023), represents the most significant gap in the variable set. Popularity operates as an independent driver of a player's value through its effect on brand image and commercial revenue generation, and its omission means that the Premier League coefficients in both Set A and Set B may be attributed to the league's large media exposure rather than a simple structural league premium only due to clubs having more money. Future research should incorporate Google Trends search volume indices or social media follower counts as explicit popularity regressors, following the approach proposed by Malagón-Selma et al. (2023).

Second, contract length remaining at the time of transfer is not included in the present analysis due to data availability constraints (exact wages, salaries and contract clauses are not publicly available information and are often not sourced or differ from media to media). While contract length is a recognised determinant of transfer fees, players whose contracts are expiring in one year are transferred for significantly lower fees than players with three or more years remaining, it does not affect market value itself, since a player's contractual situation has no bearing on their on-pitch characteristics, performance output, or perceived quality. The omission therefore concerns Set B exclusively: contract length is likely to explain a share of the unexplained fee-MV variation, but its absence does not introduce bias into the Set A market value regressions.

Third, the study's design does not permit causal inference. The regression coefficients should be interpreted as conditional associations rather than causal effects.

Fourth, the sample covers only permanent fee-paying transfers to clubs in the Top 5 leagues. The generalisability of the findings to transfers involving selling clubs outside these leagues, or to transfers within lower divisions, cannot be assumed without further investigation. Poli et al. (2024) find broadly consistent results across a global sample of 54 national associations, which provides some external reassurance, though the specific magnitudes of the league premium, the forward premium, and the age effect may differ in lower-profile markets.

Finally, manual verification of a small number of observations revealed at least one instance of a name collision in the data merge procedure, where a transfer fee from a lower-profile player was incorrectly attributed to a high-profile defender of the same name. This suggests that a small residual number of mismatched observations may persist in the final sample despite the deduplication step applied during cleaning. Given the scale of the dataset (789 regression

observations) and the magnitude of the coefficients estimated, this residual data quality issue is not material to the substantive findings, but it is acknowledged as a limitation.

6. Conclusion

This study set out to answer two related but distinct questions about the European football transfer market: which player characteristics systematically determine Transfermarkt market value, and which factors explain when and why realised transfer fees deviate from those market value estimates. Drawing on 795 permanent fee-paying transfers across the five major European leagues over the summer windows of 2021, 2022, and 2023, and employing Principal Component Analysis to handle the multicollinearity inherent in performance data followed by two parallel progressive OLS regression frameworks, the study provides empirically grounded answers to both.

On the first question, market value is explained to a substantial degree ($R^2 = 0.515$) by a structured combination of performance metrics, age, position, and league of destination. Age emerges as the dominant variable: each additional year is associated with an approximately 13.5% reduction in market value, confirming the youth premium documented across the literature (Müller et al., 2017; Franceschi et al., 2024) and consistent with Rosen's (1981) superstar framework. The three principal components of performance are all highly significant, with attacking output (PC1) rewarded and creative involvement (PC2) associated with lower market values after controlling for goal-scoring output. Positional asymmetries are pronounced: the forward subgroup achieves $R^2 = 0.515$, compared to $R^2 = 0.229$ for defenders, reflecting the fundamental limitation that the publicly available performance variables in football capture attacking contribution far more comprehensively than defensive ones, making it difficult to truly assess a defender's quality. The Premier League commands a market value premium of approximately 158% over the Bundesliga baseline, dominating all other league effects and extending the English premium findings of Bell et al. (2024).

On the second question, the systematic component of the deviation between realised fees and market value is more modest in magnitude ($R^2 = 0.103$). Age exerts a positive and significant effect on the fee-market value gap, indicating that buying clubs discount age less aggressively than the Transfermarkt community, consistent with the work of Carmichael & Thomas (1993), in which clubs value the reduced uncertainty associated with proven players. Forwards command fees approximately 26% above their market values, a systematic positional premium that aligns with the goal-scoring scarcity premium documented by Majewski (2016). Premier League clubs pay fees approximately 65% above the market value baseline, producing a cumulative effect in which English transfers are priced at substantial premium both for market value and transfer fee. Injury days is positively and significantly associated with the fee-MV

gap, a counterintuitive result at first but understood as a selection effect, where the most heavily used, tracked and highest-profile players accumulate both more extensive injury records and larger fee-MV premia.

The study contributes to the empirical literature on football player valuation in several ways. First, it provides one of the few multi-league analyses of the post-pandemic transfer market covering the 2021–2023 summer windows, addressing the temporal gap identified by Franceschi et al. (2024) in their systematic review. Second, it applies Principal Component Analysis to address multicollinearity in performance data, a methodological approach advocated by Kirschstein & Liebscher (2019) but rarely used in transfer fee models. Third, it introduces the two-stage analytical framework of modelling market value determinants and fee-MV deviations separately, providing a cleaner treatment of the crowd-sourced valuation signal than standard approaches that treat market value as another regressor. Fourth, it introduces injury history as an explicit predictor and documents its position-dependent and stage-dependent effects. Fifth, it quantifies the Premier League structural premium at two distinct stages of the price formation process, providing the most comprehensive evidence to date of the English league's systematic advantage over other Top 5 leagues in the recent transfer market era.

Future research extending this framework or results should prioritise three directions. Most importantly, the integration of a social media popularity variable, following the Google Trends methodology proposed by Malagón-Selma et al. (2023), would allow the direct testing of Rosen's (1981) superstar premium in the market value and fee-MV contexts. Second, the inclusion of contract length remaining at the time of transfer would capture the bargaining power dimension of fee determination that multiple studies identify as one of the strongest structural predictors of fee-MV deviations (Carmichael et al., 1999; Poli et al., 2022). Third, an extension of the sample to include non-Top 5 leagues, following the global approach of Poli et al. (2024), would allow the league premium findings to be contextualised within a broader international market structure.

The football transfer market represents one of the most publicly observable and economically significant labour markets in the world. Total global transfer spending reached €12.2 billion in 2023 alone, and the decisions made by clubs, agents, and governing bodies shape the financial sustainability of football institutions, the careers of thousands of professional athletes, and the competitive balance of competitions watched by hundreds of millions of people. Understanding how the market forms its valuations, and how actual fees deviate from those valuations, is therefore not merely an academic exercise but a practical

consideration for every stakeholder in the sport. This study demonstrates that the Transfermarkt market valuation and the realised transfer fee are distinct but related pricing signals, each driven by a structured combination of player characteristics and market context, and that departures between the two are themselves systematic and explainable rather than arbitrary. The two-stage framework developed here may serve as a foundation for future research into football player valuation that treats the crowd-sourced valuation and the realised transaction as complementary objects of study rather than substitutes.

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Appendixes

Appendix A. Trimmed Sample Robustness Checks.

	Dependent variable:	
	log_mv	fee_mv_gap
	Set A: log(Market Value)	Set B: log(Fee) - log(MV)
	(1)	(2)
PC1 (Attacking Output)	0.220****	-0.046**
	(0.022)	(0.019)
PC2 (Creative Involvement)	-0.177****	0.069***
	(0.023)	(0.021)
PC3 (Residual)	-0.271****	0.002
	(0.037)	(0.033)
Age	-0.107****	0.044****
	(0.009)	(0.007)
Midfielder (ref: Defender)	-0.224***	0.053
	(0.074)	(0.065)
Forward (ref: Defender)	-0.453****	0.265****
	(0.086)	(0.074)
Injury Days	-0.0003	0.001***
	(0.0004)	(0.0004)
Season 2023	0.040	-0.050
	(0.069)	(0.060)
Season 2024	0.063	-0.024
	(0.067)	(0.059)
La Liga	-0.051	0.037
	(0.097)	(0.087)
Ligue 1	-0.101	0.169**
	(0.088)	(0.077)
Premier League	0.885****	0.245***
	(0.086)	(0.077)
Serie A	0.015	0.203***
	(0.085)	(0.075)
Constant	18.808****	-1.626****
	(0.226)	(0.190)
Observations	724	709
R^2	0.424	0.120
Adjusted R^2	0.414	0.104

*Note: Standard errors in parentheses. Reference categories: Bundesliga (league), Defender (position), 2022 (season). Sample sizes: Set A n=724, Set B n=709. Significance levels: *p<0.1; *p<0.05; *p<0.01; *p<0.001. Source: Author's own calculations.*

Appendix B. Position-Specific Fee-Market Value Gap Regressions.

	Dependent variable:		
	log(Fee) - log(Market Value)		
	Defenders	Midfielders	Forwards
	(1)	(2)	(3)
PC1 (Attacking Output)	0.008	-0.016	-0.036
	(0.076)	(0.067)	(0.025)
PC2 (Creative Involvement)	0.085	0.020	0.081**
	(0.052)	(0.068)	(0.032)
PC3 (Residual)	-0.040	-0.032	0.072
	(0.080)	(0.084)	(0.060)
Age	0.00004	0.077****	0.034***
	(0.015)	(0.016)	(0.013)
Injury Days	0.003****	0.001	0.001
	(0.001)	(0.001)	(0.001)
Season 2023	-0.113	-0.052	-0.018
	(0.118)	(0.145)	(0.115)
Season 2024	-0.053	-0.081	0.005
	(0.122)	(0.140)	(0.110)
Constant	-0.454	-2.299****	-1.034***
	(0.381)	(0.400)	(0.332)
Observations	271	217	301
R^2	0.077	0.112	0.055
Adjusted R^2	0.053	0.083	0.033

*Note: Standard errors in parentheses. Reference category: 2022 (season). Sample sizes: Defenders n=271, Midfielders n=217, Forwards n=301. Significance levels: *p<0.1; *p<0.05; *p<0.01; *p<0.001. Source: Author's own calculations.*

Appendix C. Variance Inflation Factor Diagnostics.

This appendix presents the Variance Inflation Factors (VIF) for the full pooled regression specifications used in Sets A and B. Because Model A4 and Model B4 share an identical set of right-hand-side regressors, their VIF values are mathematically identical and are therefore reported in a single table. For factor variables with multiple levels (season_year and league), the generalized VIF (GVIF) and the scale-adjusted measure GVIF^(1/(2·Df)) are reported, where the latter is comparable to standard VIF values for continuous variables. All VIF values are well below the conventional threshold of 5, indicating that the estimated coefficients are not destabilized by multicollinearity.

Variable	GVIF	Df	GVIF^(1/(2·Df))
PC1 (Attacking Output)	1.948	1	1.396
PC2 (Creative Involvement)	1.066	1	1.032
PC3 (Residual)	1.180	1	1.086
Age	1.085	1	1.042
pos_MF	1.517	1	1.232
pos_FW	2.357	1	1.535
injury_days	1.033	1	1.016
season_year	1.038	2	1.009
league	1.092	4	1.011

Note: VIF values are identical for Models A4 and B4 because both share the same explanatory variables. The GVIF^(1/(2·Df)) column provides the comparable VIF measure for factor variables (season_year and league) and is interpreted on the same scale as standard VIF. All values are well below the conventional threshold of 5, confirming the absence of problematic multicollinearity. Source: Author's own calculations using the car package (Fox & Weisberg, 2019) in R.

Appendix D. Residual Diagnostic Plots.

This appendix presents the standard residual diagnostic plots for the two full pooled regression specifications, Model A4 (Determinants of Market Value) and Model B4 (Fee-Market Value Gap). Each set of plots includes (i) residuals versus fitted values, used to detect non-linearity and heteroscedasticity; (ii) a normal Q-Q plot, used to assess the normality of residuals; (iii) a scale-location plot, used to detect heteroscedasticity; and (iv) residuals versus leverage, used to identify influential observations. Together, these diagnostics confirm that the OLS assumptions are reasonably met for both models.

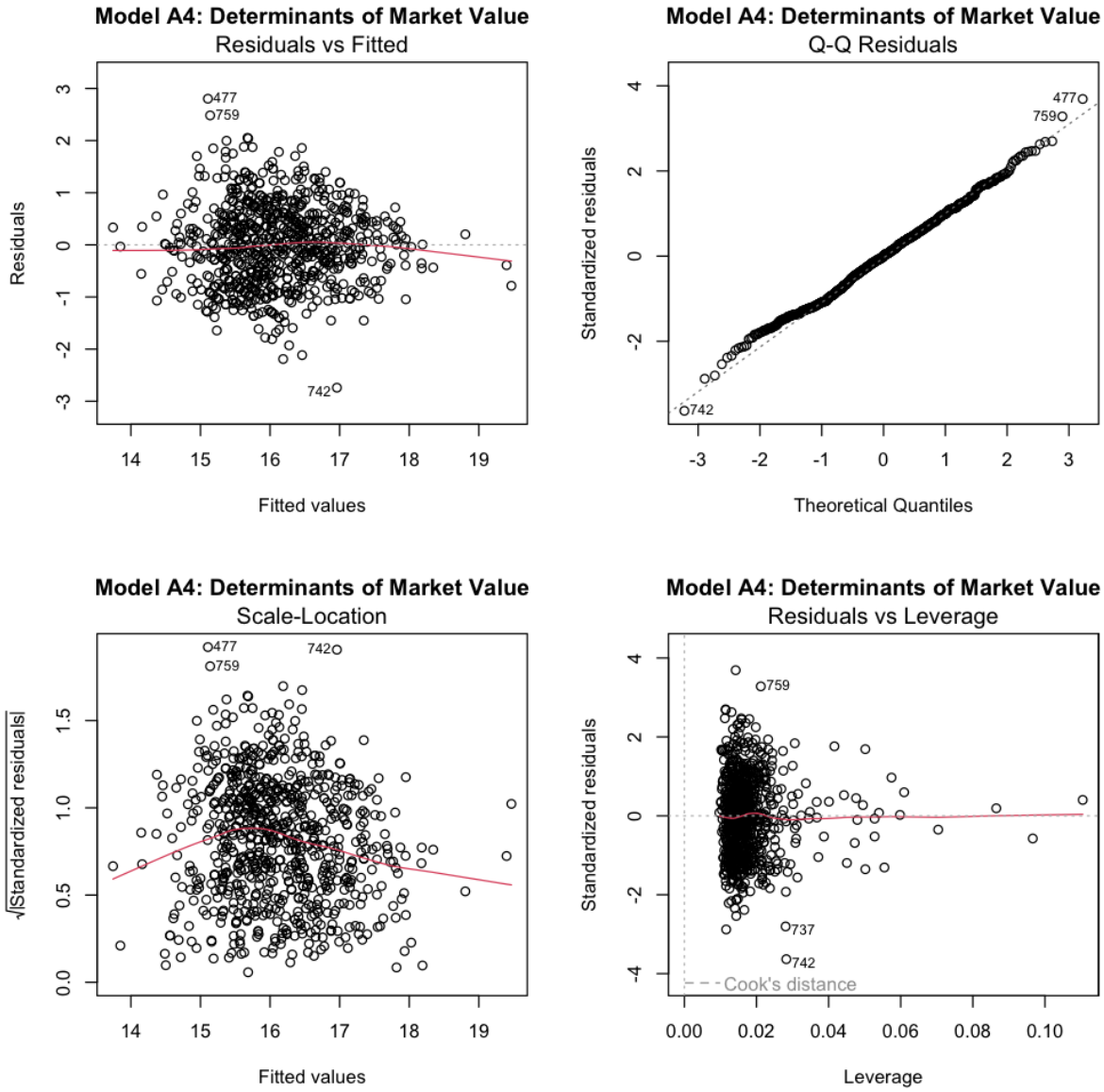


Figure D1. Residual diagnostic plots for Model A4 (Determinants of Market Value).

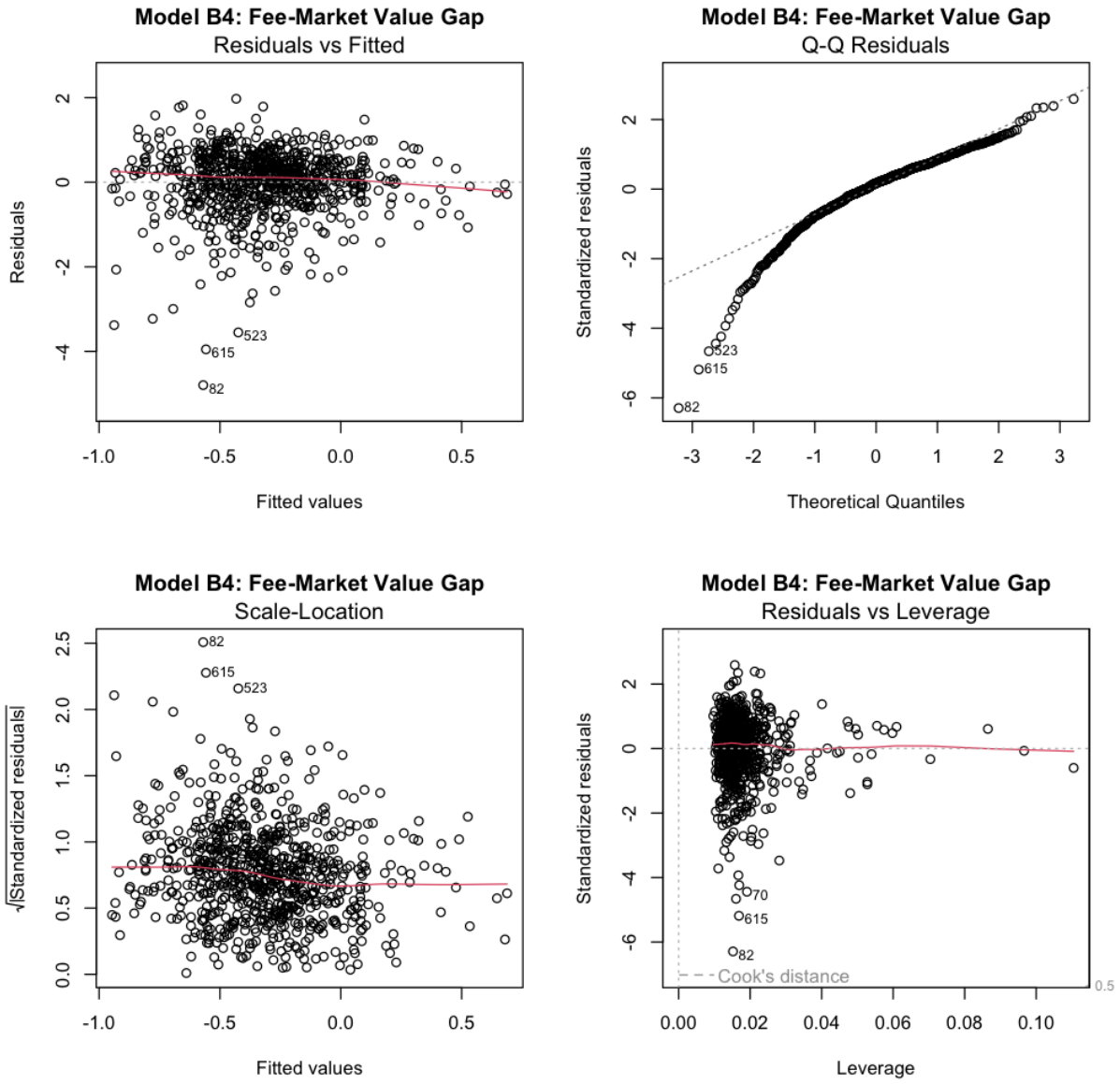


Figure D2. Residual diagnostic plots for Model B4 (Fee-Market Value Gap).

The residuals versus fitted plots show no obvious pattern, indicating linearity and homoscedasticity hold reasonably well. The Q-Q plots show the residuals follow an approximately normal distribution, with mild deviation in the tails consistent with the inherent skewness of transfer market data. The scale-location plots confirm constant variance across the fitted range. Finally, the residuals versus leverage plots indicate no observations exceed the Cook's distance threshold of concern, confirming that no individual data point exerts disproportionate influence on the estimated coefficients.

Appendix E. R Analysis Script.

This appendix presents the full R analysis script used to produce the empirical results reported in Chapters 3 and 4. The script reproduces the data preparation, Principal Component Analysis, all regression specifications (Sets A and B), position-specific subgroup regressions, robustness checks, and diagnostic outputs. The dataset `(transferred_pca.csv)` is required as the input file. All analyses were conducted in R version 4.x using the `dplyr`, `stargazer`, `car`, and `ggplot2` packages.

```
# ============================================================
# THESIS ANALYSIS SCRIPT
# Determinants of Market Value and Transfer Fee Deviations
# in European Football: A Two-Stage Quantitative Analysis
# of Player Pricing in the Top 5 Leagues
#
# Author: Victor Auberger
# ESCP Business School — BSc in Management
# May 2026
# ============================================================

# Load required packages
library(dplyr)
library(tidyr)
library(stargazer)
library(car)
library(ggplot2)
library(gridExtra)

# ============================================================
# DATA LOADING AND PREPARATION
# ============================================================

# Load the cleaned and pre-processed analytical dataset
transferred_pca <- read.csv("Data/transferred_pca.csv")
```

```
# Create transformed variables for regression analysis
transferred_pca <- transferred_pca %>%
  mutate(
    log_mv = log(market_value_in_eur),
    fee_mv_gap = log_transfer_fee - log(market_value_in_eur)
  )

# Convert categorical variables to factors for fixed effects
transferred_pca$season_year <-
as.factor(transferred_pca$season_year)
transferred_pca$league <- as.factor(transferred_pca$league)

# ============================================================
# PRINCIPAL COMPONENT ANALYSIS
# ============================================================

# Construct per-90-minute performance variables
mins_per90 <- transferred_pca$Min_Playing / 90
perf_vars <- data.frame(
  Gls_per90  = transferred_pca$Gls / mins_per90,
  Ast_per90  = transferred_pca$Ast / mins_per90,
  xG_per90   = transferred_pca$xG_Expected / mins_per90,
  npxG_per90 = transferred_pca$npxG_Expected / mins_per90,
  PrgC_per90 = transferred_pca$PrgC_Progression / mins_per90,
  PrgP_per90 = transferred_pca$PrgP_Progression / mins_per90
) %>% na.omit()

# Run PCA with standardisation
pca_result <- prcomp(perf_vars, center = TRUE, scale. = TRUE)
summary(pca_result)
print(pca_result$rotation)

# ============================================================
```

```
# SET A: DETERMINANTS OF MARKET VALUE
# ============================================================


# Pooled progressive models
model_A1 <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + season_year,
               data = transferred_pca)
model_A2 <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW + season_year,
               data = transferred_pca)
model_A3 <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW +
                 injury_days + season_year,
               data = transferred_pca)
model_A4 <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW +
                 injury_days + season_year + league,
               data = transferred_pca)


# Position-specific subgroup regressions
model_A_DF <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + injury_days
+ season_year,
                 data = filter(transferred_pca, pos_simple ==
"DF"))
model_A_MF <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + injury_days
+ season_year,
                 data = filter(transferred_pca, pos_simple ==
"MF"))
model_A_FW <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + injury_days
+ season_year,
                 data = filter(transferred_pca, pos_simple ==
"FW"))


# Trimmed sample robustness check (5%-95% market value)
mv_bounds <- quantile(transferred_pca$market_value_in_eur,
                      c(0.05, 0.95), na.rm = TRUE)
trimmed_A <- transferred_pca %>%
```

```
  filter(market_value_in_eur >= mv_bounds[1] &
         market_value_in_eur <= mv_bounds[2])
model_A_trim <- lm(log_mv ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW +
                     injury_days + season_year + league,
                   data = trimmed_A)


# ============================================================
# SET B: FEE-MARKET VALUE GAP
# ============================================================


# Pooled progressive models
model_B1 <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age +
season_year,
               data = transferred_pca)
model_B2 <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW + season_year,
               data = transferred_pca)
model_B3 <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW +
                 injury_days + season_year,
               data = transferred_pca)
model_B4 <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age + pos_MF +
pos_FW +
                 injury_days + season_year + league,
               data = transferred_pca)


# Position-specific subgroup regressions
model_B_DF <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age +
injury_days + season_year,
                 data = filter(transferred_pca, pos_simple ==
"DF"))
model_B_MF <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age +
injury_days + season_year,
                 data = filter(transferred_pca, pos_simple ==
"MF"))
```

```
model_B_FW <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age +
injury_days + season_year,
                 data = filter(transferred_pca, pos_simple ==
"FW"))


# Trimmed sample robustness check (5%-95% transfer fee)
fee_bounds <- quantile(transferred_pca$transfer_fee,
                       c(0.05, 0.95), na.rm = TRUE)
trimmed_B <- transferred_pca %>%
  filter(transfer_fee >= fee_bounds[1] & transfer_fee <=
fee_bounds[2])
model_B_trim <- lm(fee_mv_gap ~ PC1 + PC2 + PC3 + Age + pos_MF
+ pos_FW +
                     injury_days + season_year + league,
                   data = trimmed_B)


# ============================================================
# DIAGNOSTIC TESTS
# ============================================================


# Variance Inflation Factors
vif_A4 <- vif(model_A4)
vif_B4 <- vif(model_B4)
print(vif_A4)
print(vif_B4)


# Residual diagnostic plots
par(mfrow = c(2, 2))
plot(model_A4)
plot(model_B4)
par(mfrow = c(1, 1))


# ============================================================
# DESCRIPTIVE STATISTICS AND FIGURES
```

```
# ============================================================


# Sample distribution by league and season
table_distribution <- transferred_pca %>%
  mutate(transfer_year = as.numeric(as.character(season_year)) - 1) %>%
  group_by(league, transfer_year) %>%
  summarise(n = n(), .groups = "drop") %>%
  pivot_wider(names_from = transfer_year, values_from = n, values_fill = 0)
print(table_distribution)


# Correlation matrix
cor_matrix <- transferred_pca %>%
  mutate(log_fee = log_transfer_fee, log_mv = log(market_value_in_eur)) %>%
  select(log_fee, PC1, PC2, PC3, Age, log_mv, injury_days) %>%
  cor(use = "complete.obs") %>%
  round(3)
print(cor_matrix)


# Scree plot for PCA
var_explained <- data.frame(
  PC = paste0("PC", 1:6),
  Variance = round(summary(pca_result)$importance[2,] * 100, 1)
)


ggplot(var_explained, aes(x = PC, y = Variance, group = 1)) +
  geom_line(colour = "steelblue", linewidth = 1) +
  geom_point(colour = "steelblue", size = 3) +
  geom_hline(yintercept = 100/6, linetype = "dashed", colour = "grey50") +
  labs(title = "Scree Plot — Principal Component Analysis",
```

```
       x = "Principal Component", y = "Variance Explained
(%)") +
  theme_minimal()


# Transfer fee distribution
p1 <- ggplot(transferred_pca, aes(x = transfer_fee/1e6)) +
  geom_histogram(bins = 40, fill = "steelblue", colour =
"white", alpha = 0.8) +
  labs(title = "Raw Transfer Fee Distribution",
       x = "Transfer Fee (€ millions)", y = "Frequency") +
  theme_minimal()


p2 <- ggplot(transferred_pca, aes(x = log_transfer_fee)) +
  geom_histogram(bins = 40, fill = "steelblue", colour =
"white", alpha = 0.8) +
  labs(title = "Log-Transformed Transfer Fee Distribution",
       x = "log(Transfer Fee)", y = "Frequency") +
  theme_minimal()


grid.arrange(p1, p2, ncol = 2)


# ============================================================
# END OF SCRIPT
# ============================================================
```